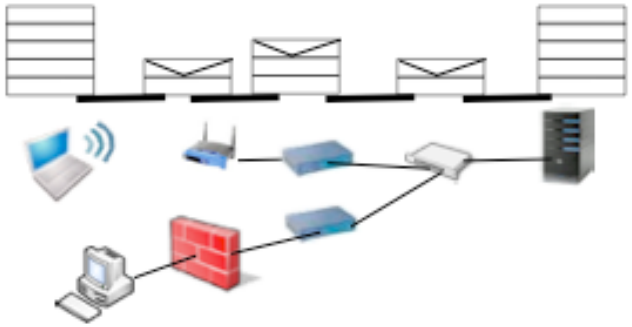
Computer Networking

Principles
Protocols
and
Practice



Week 3 : MultiProtocol Label Switching

Olivier Bonaventure
http://inl.info.ucl.ac.be/

MPLS
MultiProtocol Label Switching

- Outline
  - Multiprotocol Label Switching
    - The label swapping forwarding paradigm
    - Integrating label swapping and IP
  - Utilisations of MPLS
    - Destination based packet forwarding
    - Simpler ISP backbones
    - Traffic engineering
    - QoS support
    - Fast restoration

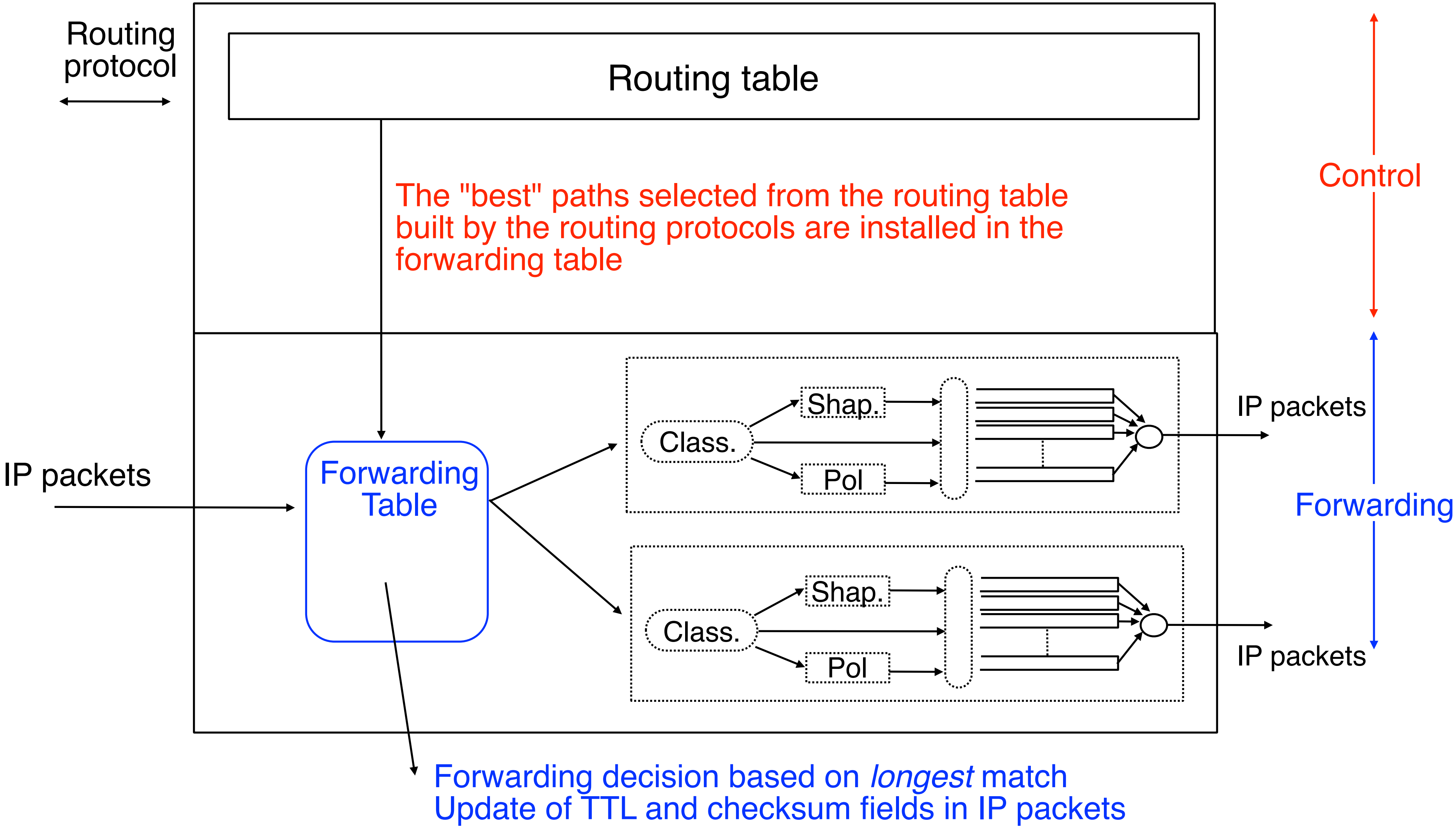
High performance packet switching

- The challenge
  - To efficiently support high speed networks, a router should be capable of switching and forwarding packets at line rate on all interfaces
  - Inter-packet time on high-speed interfaces

Inter-packet time in microsec versus packet size in bytes

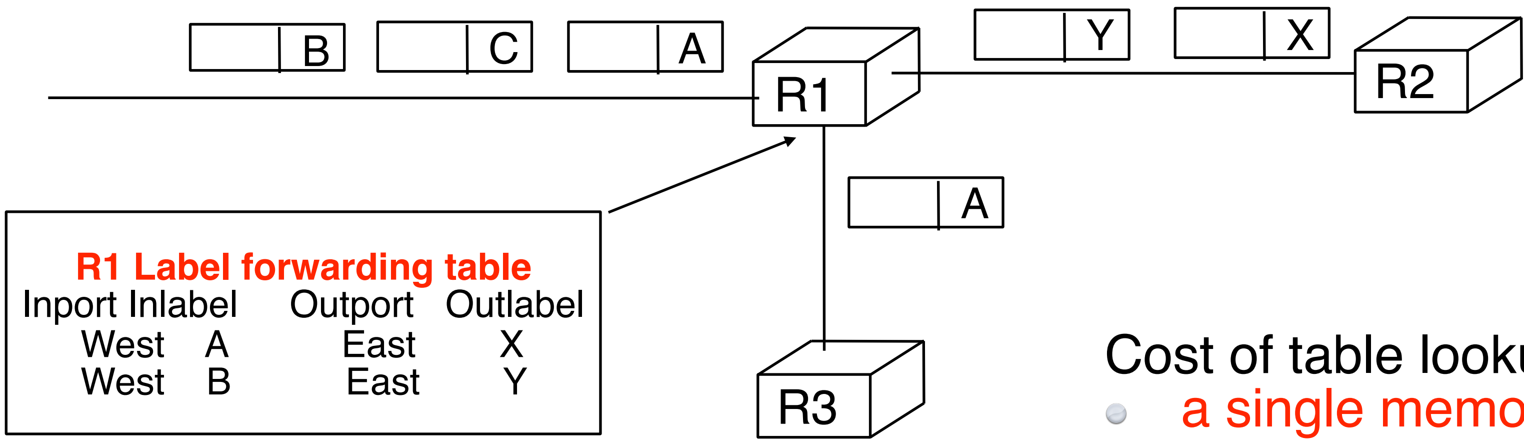
Interface	40 bytes	250 bytes	1500 bytes
10 Mbps	32	200	1200
100 Mbps	3,2	20	120
155 Mbps	2,06	12,9	77,42
622 Mbps	0,51	3,22	19,29
2.4 Gbps	0,13	0,83	5
10 Gbps	0,03	0,2	1,2

Architecture of a normal IP router



Label swapping
A simpler forwarding paradigm

- Principle
  - On packet arrival, router analyses
    - Packet Label
    - *Input Port*
  - Based on label forwarding label, router decides
    - Output Port for outgoing packet
    - Packet Label for outgoing packet



R1 Label forwarding table

Inport	Inlabel	Outport	Outlabel
West	A	East	X
West	B	East	Y

Cost of table lookup
- a single memory access !

Historical perspective

- Early 1990's
  - IP routers were essentially high-speed computers with special software
    - Impossible to implement IP forwarding in hardware
    - Performance of routers was limited by their CPU
      - forwarding assistance with caches to avoid expensive route lookups
  - Asynchronous Transfer Mode
    - Emerging technology relying on label swapping
    - Hardware implementation was easy at high speeds
      - 155 Mbps, 622 Mbps
    - To support voice (telephony) all packets were divided in 48 bytes cells with 5 bytes header for label

Historical perspective (2)

- Early 2000's
  - Moore's Law
    - CPU performance doubles every 18 months
    - VLSI integration allows more complex CPUs
  - IP routers
    - New advances in forwarding algorithms
      - IP forwarding can be easily performed in hardware
      - chips are complex but wire speed at 2.4 Gbps or 10 Gbps works
    - Label swapping is not anymore necessary from performance or hardware implementation point of view
  - ATM
    - 48+5 bytes cell size is major drawback and implementation cost
    - ATM interfaces on IP routers do not exist above 622 Mbps
    - ATM mainly restricted to medium speed ADSL->1Gbps

Historical perspective (3)

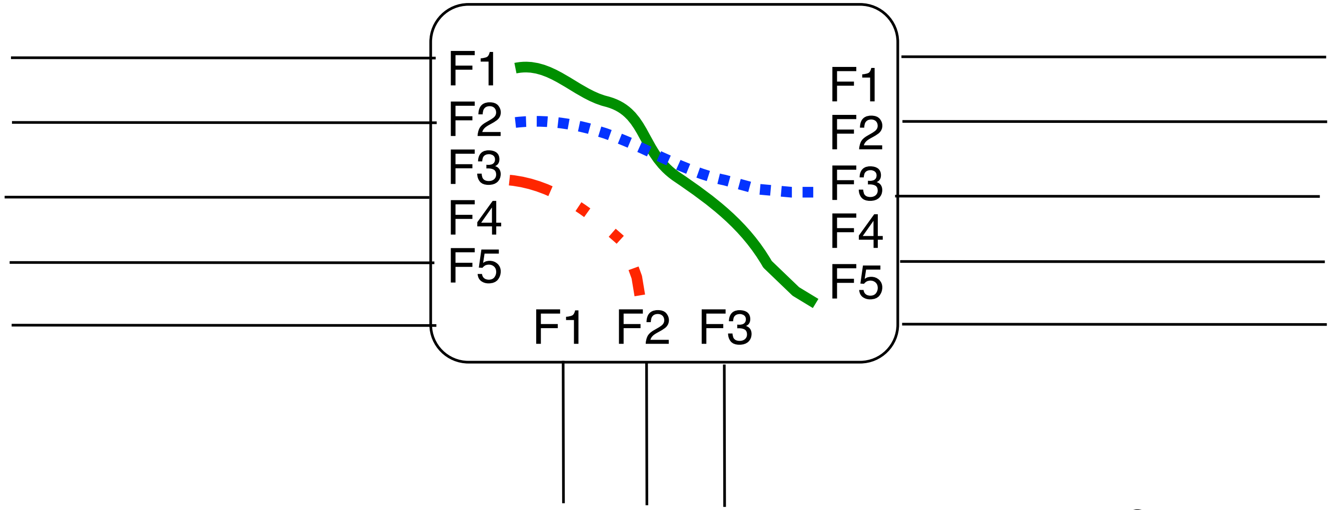
- Current motivations for MPLS
  - Applications
    - destination based routing
    - traffic engineering
    - QoS
    - fast restoration
    - Virtual Private Networks
  - Utilisation of MPLS to control optical or transmission devices close to label swapping
    - fibre switch
    - lambda switch
    - TDM (SONET/SDH) switches

Historical perspective (4)

- MPLS controlled devices
  - MPLS and IP routing are used to establish flows through these devices
  - equipment uses a special forwarding mechanism
- Fibre switch

Fiber switching table

Inport	InFiber	OutPort	OutFiber
West	F1	East	F5
West	F2	East	F3
West	F3	South	F2

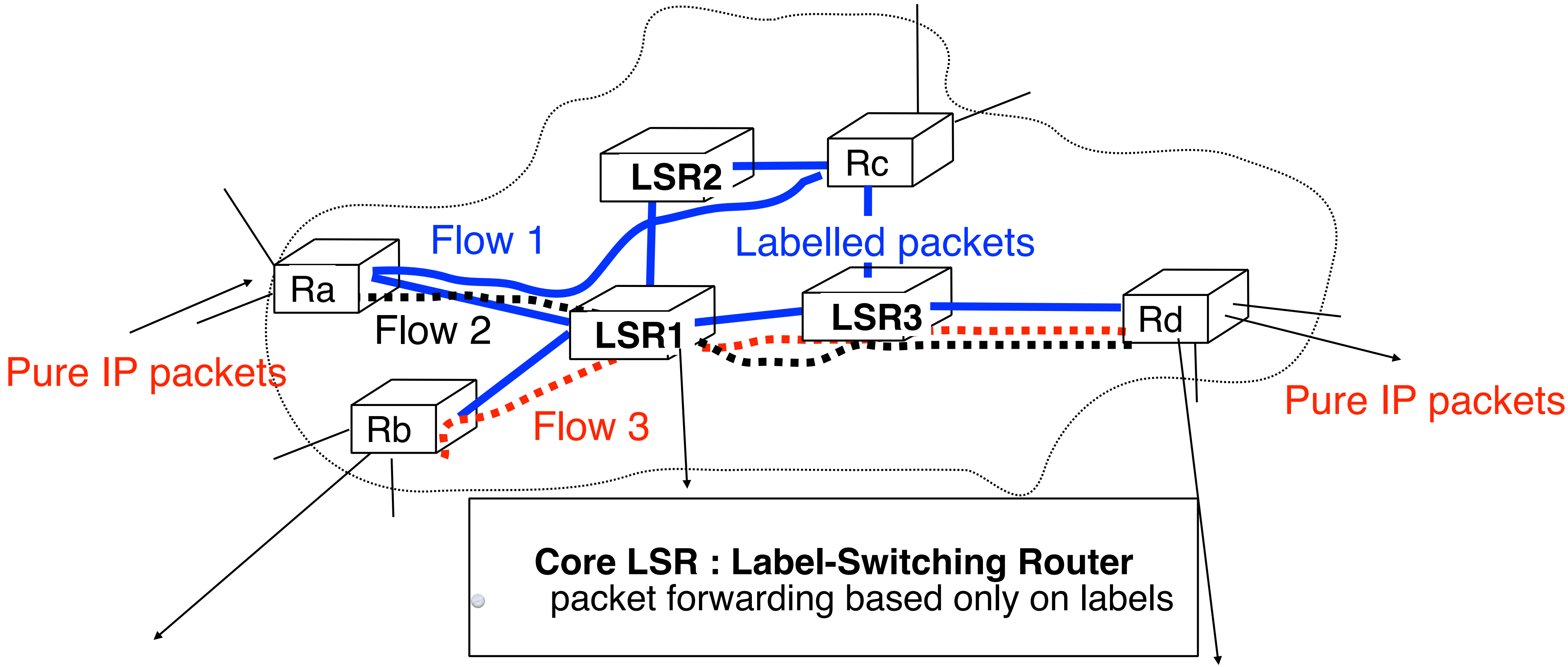


- λ switch

MPLS
MultiProtocol Label Switching

- Plan
  - Multiprotocol Label Switching
    - The label swapping forwarding paradigm
    - Integrating label swapping and IP
  - Utilisation of MPLS
    - Destination based packet forwarding
    - Simpler ISP backbones
    - Traffic engineering
    - QoS support
    - Fast restoration

Integrating label swapping and IP



Core LSR : Label-Switching Router
- packet forwarding based only on labels

Ingress Edge Router
- inserts labels on packets sent through backbone

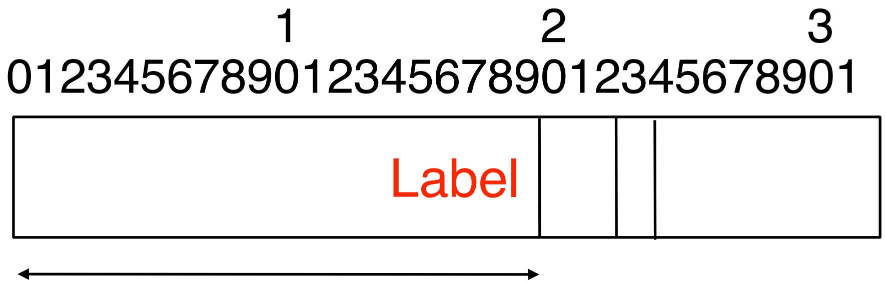
Egress Edge router
- removes labels before sending packets outside MPLS network

Integrating label swapping and IP (2)

- Problems to be solved
  - What is a labelled IP packet ?
  - What is the behaviour of a core LSR ?
  - What is the behaviour of an edge LSR ?

Labelled IP packets

- Generic solution
  - Insert special 32 bits header in front of IP packet

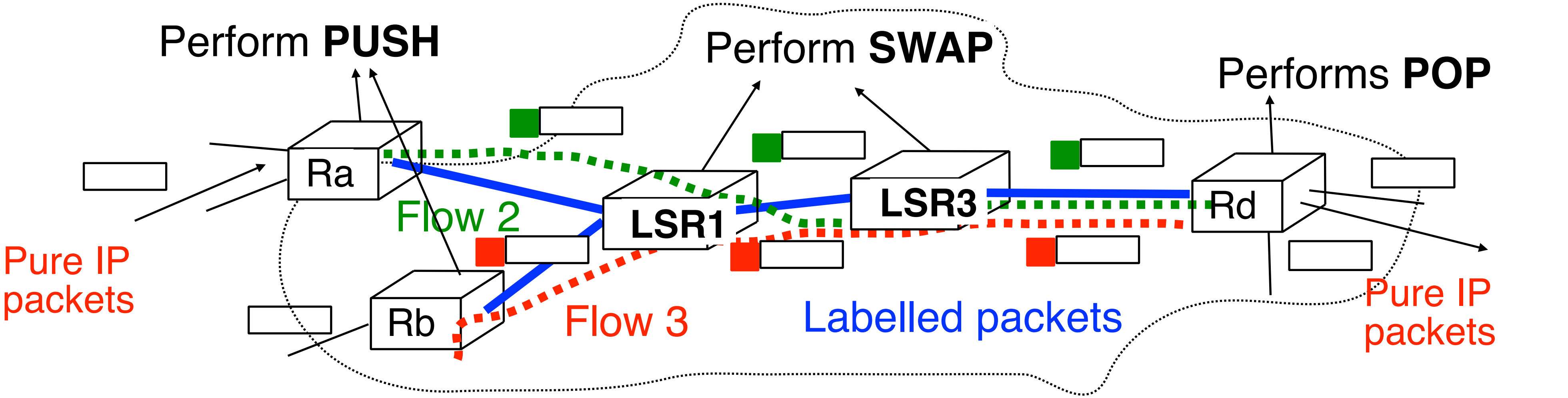


20 bits :1048576 different labels

- In the past : Technology specific solutions
  - Reuse the already available "labels" below layer 3
    - Frame Relay
    - Asynchronous Transfer Mode
    - Fibre/lambda switching with special label semantics
    - SONET/SDH with special label semantics

Operations performed on labelled packet

- Three types of operations
  - PUSH
    - insert a label in front of a received packet
  - SWAP
    - change the value of the label of a received labelled packet
  - POP
    - remove the label in front of a received labelled packet



Content of the Label forwarding table

InLabel NextHop; Op

Identification of the label of the incoming packet

- Operation to be performed on label
  - swap label in front of packet
  - push new label in front of packet
  - pop label in front of packet

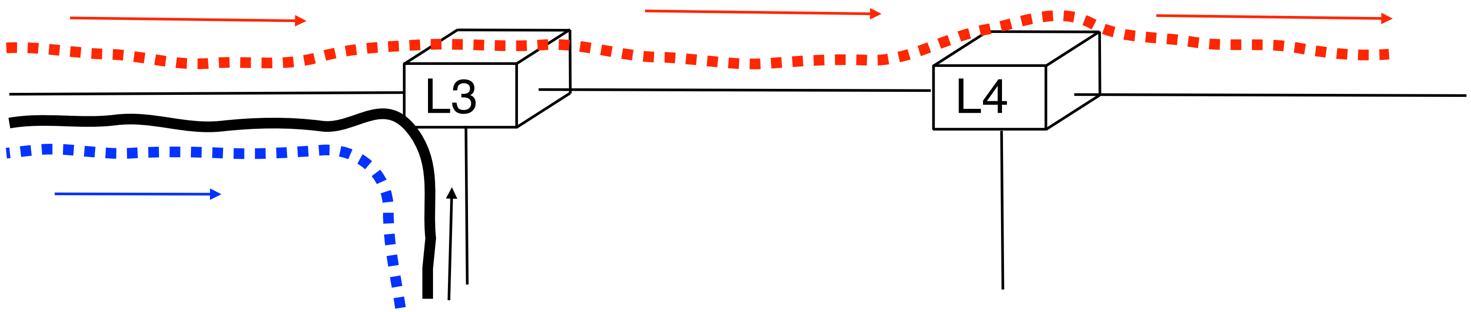
- Next hop for the packet
  - outgoing interface
    - packet sent to another LSR
  - LSR itself
    - packet destination is LSR
    - packet needs to be routed as a normal IP packet after pop

Per interface Label space

- A LSR may manage its labels with 2 methods
  - Per interface label space
    - one label space for each interface
    - 2^{20} distinct labels for each interface of the LSR

L3's per-interface Label forwarding table

Inlabel	Outport	Operation
West:L3	South	SWAP(L0)
West:L4	East	SWAP(L3)
South:L3	West	SWAP(L5)



Per LSR Label space

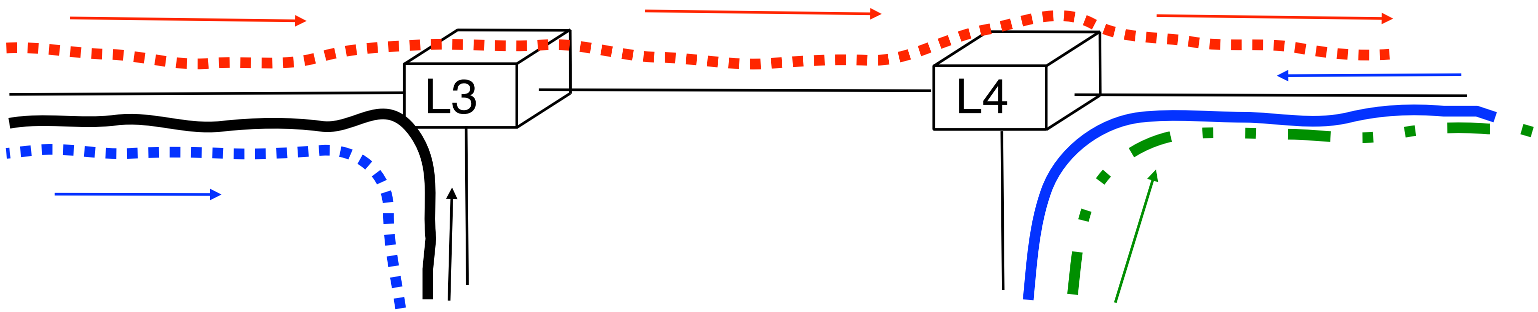
- A LSR may manage its labels with 2 methods
  - Per interface label space
    - one label space for each interface
    - 2^{20} distinct labels for each interface of the LSR

L3's per-interface Label forwarding table

Inlabel	Outport	Operation
West:L3	South	SWAP(L0)
West:L4	East	SWAP(L3)
South:L3	West	SWAP(L5)

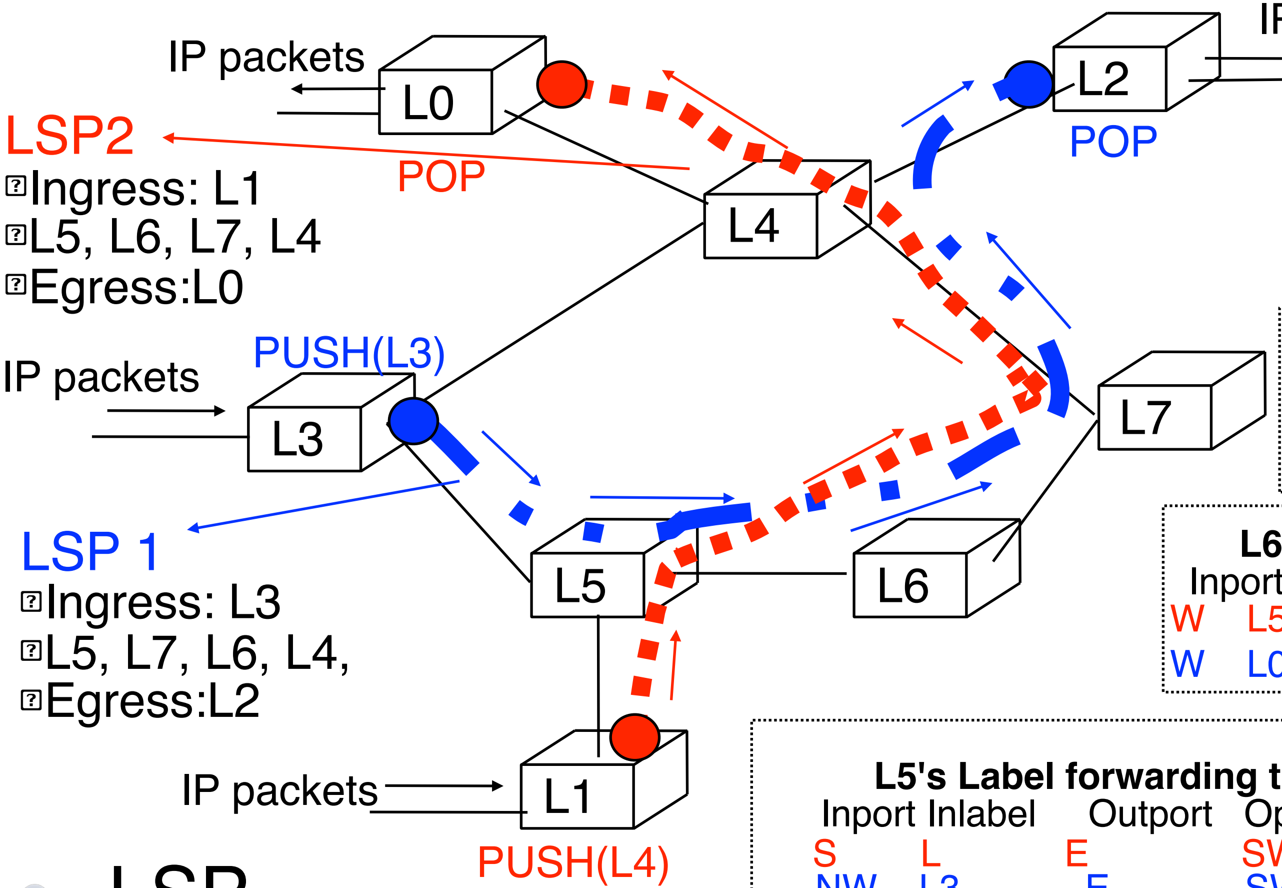
L4's per-LSR Label forwarding table

Inlabel	Outport	Operation
L0	East	SWAP(L0)
L4	South	SWAP(L3)
L3	East	SWAP(L5)



  - Per LSR label space
    - a single label space for all interfaces
    - 2^{20} distinct labels for the LSR

LSP
Label switched path



L4's Label forwarding table

Inport	Inlabel	Outport	Operation
SE	L6	NW	SWAP(L2)
SE	L1	NE	SWAP(L5)

L7's Label forwarding table

Inport	Inlabel	Outport	Operation
SW	L5	NW	SWAP(L6)
SW	L0	NW	SWAP(L1)

L6's Label forwarding table

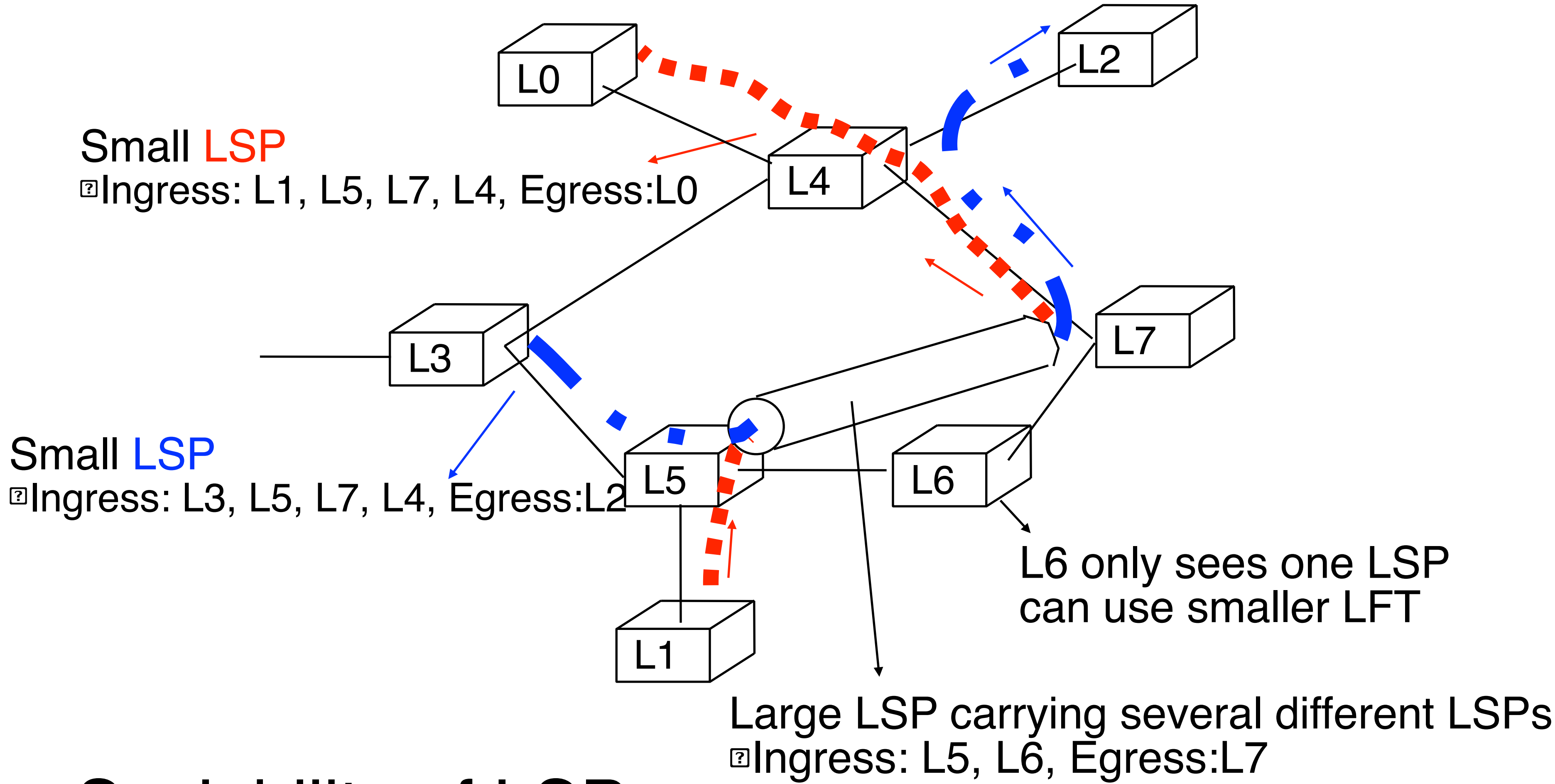
Inport	Inlabel	Outport	Operation
W	L5	NE	SWAP(L5)
W	L0	NE	SWAP(L0)

L5's Label forwarding table

Inport	Inlabel	Outport	Operation
S	L	E	SWAP(L5)
NW	L3	E	SWAP(L0)

- LSP
  - a path followed by a labelled packet over several hops starting at an ingress LSR and ending at an egress LSR
  - a LSP is usually unidirectional (ingress -> egress)

How to improve scalability ?

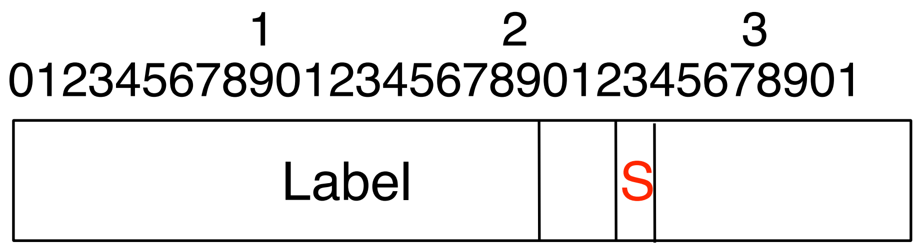


- Scalability of LSPs
  - it should be possible to place many small LSPs inside on large LSPs

Labelled IP packets (more)

- How to support hierarchy of LSPs ?
  - it should be possible to place small LSPs inside large LSPs
  - ideally, there should be no predefined limit on the number of levels supported

- Solution adopted by MPLS
  - each labelled packet can carry a stack of labels



- label at the top of the stack appears first in packet
  - S=1 if the label is at the bottom of the stack
  - S=0 if the label is not at the bottom of the stack

Content of the Label forwarding table (more)

InLabel NextHop; Operation

- Identification of the incoming label
- Interface:label for per interface label space
- label for per-LSR label space

- Next hop for the packet
  - outgoing interface
  - LSR itself
    - packet destination is LSR
    - packet needs to be

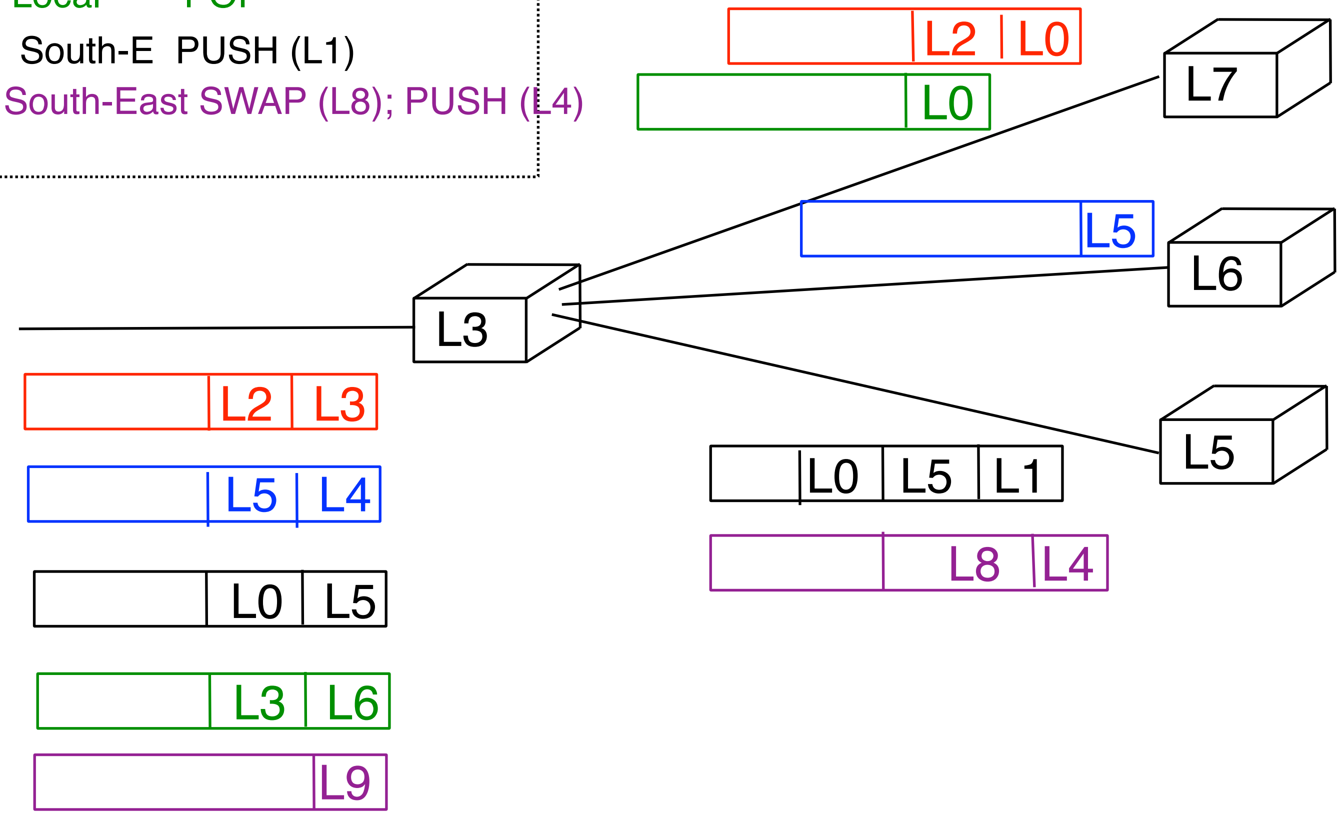
- Operation to be performed on stack
- swap label on top of the stack
- (swap label on top of the stack and) push new label on top of the stack
- pop the label stack

Content of the Label forwarding table (more) (2)

- Example

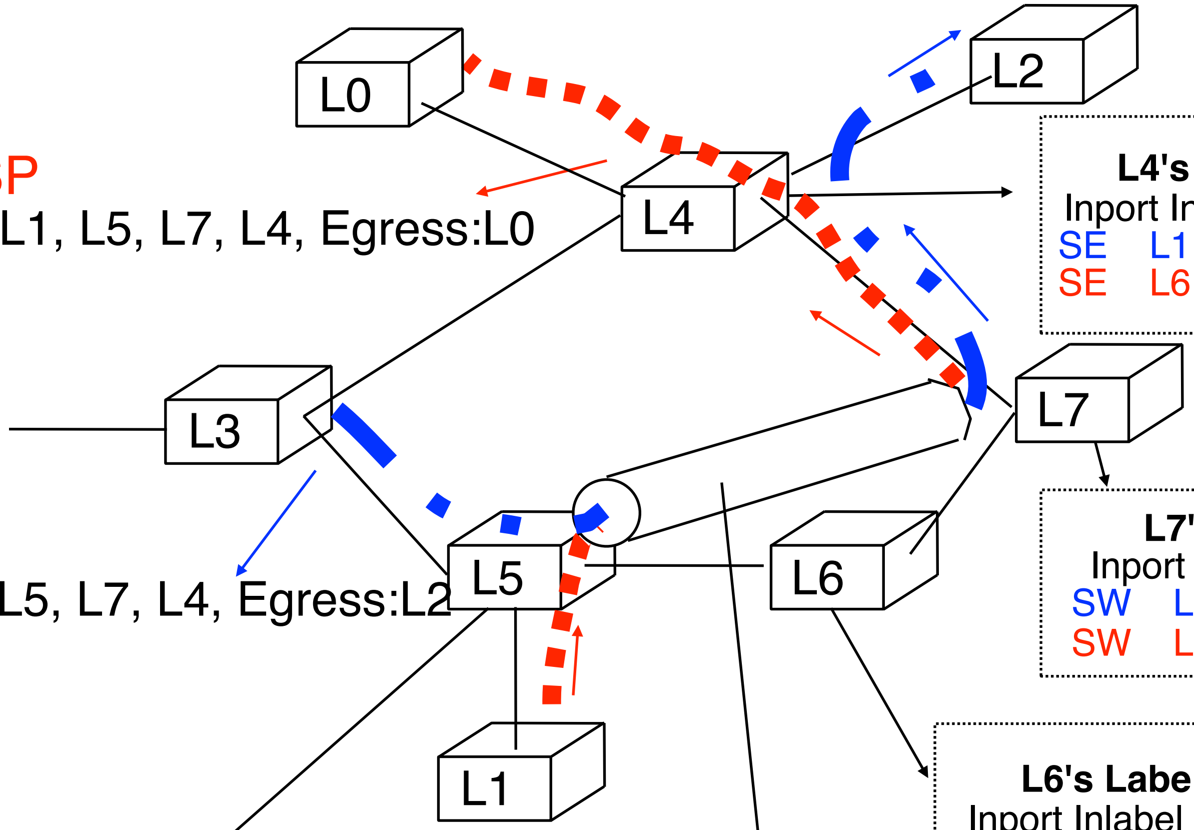
L3's Label forwarding table

Inlabel	Nexthop	Operation
L3	North-E	SWAP(L0)
L4	East	POP
L6	Local	POP
L5	South-E	PUSH (L1)
L9	South-East	SWAP (L8); PUSH (L4)



MPLS and label stacks Example

Small LSP
- Ingress: L1, L5, L7, L4, Egress:L0



Small LSP
- Ingress: L3, L5, L7, L4, Egress:L2

L4's Label forwarding table

Inport	Inlabel	Outport	Operation
SE	L1	NE	SWAP(L5)
SE	L6	NW	SWAP(L2)

L7's Label forwarding table

Inport	Inlabel	Outport	Operation
SW	L0	NW	SWAP(L1)
SW	L5	NW	SWAP(L6)

L6's Label forwarding table

Inport	Inlabel	Outport	Operation
W	L1	NE	POP

L6 only sees one LSP !

L5's Label forwarding table

Inport	Inlabel	Outport	Operation
NW	L3	E	SWAP(L0), PUSH L1
S	L4E		SWAP(L5), PUSH L1

Large LSP carrying several different LSPs
- Ingress: L5, L6, Egress:L7

Behaviour of ingress edge LSR

- How does edge LSR at ingress determine the label to be used to forward a received packet ?

  - Principle
    1. Divide the set of all possible packets into several Forwarding Equivalence Classes (FEC)
       - *a FEC is a group of IP packets that are forwarded in the same manner (e.g. over the same path, with the same forwarding treatment)*
       - examples
         - all packets sent to the same destination prefix
         - all packets sent to the same BGP next hop
    2. Associate the same label to all the packets that belong to the same FEC

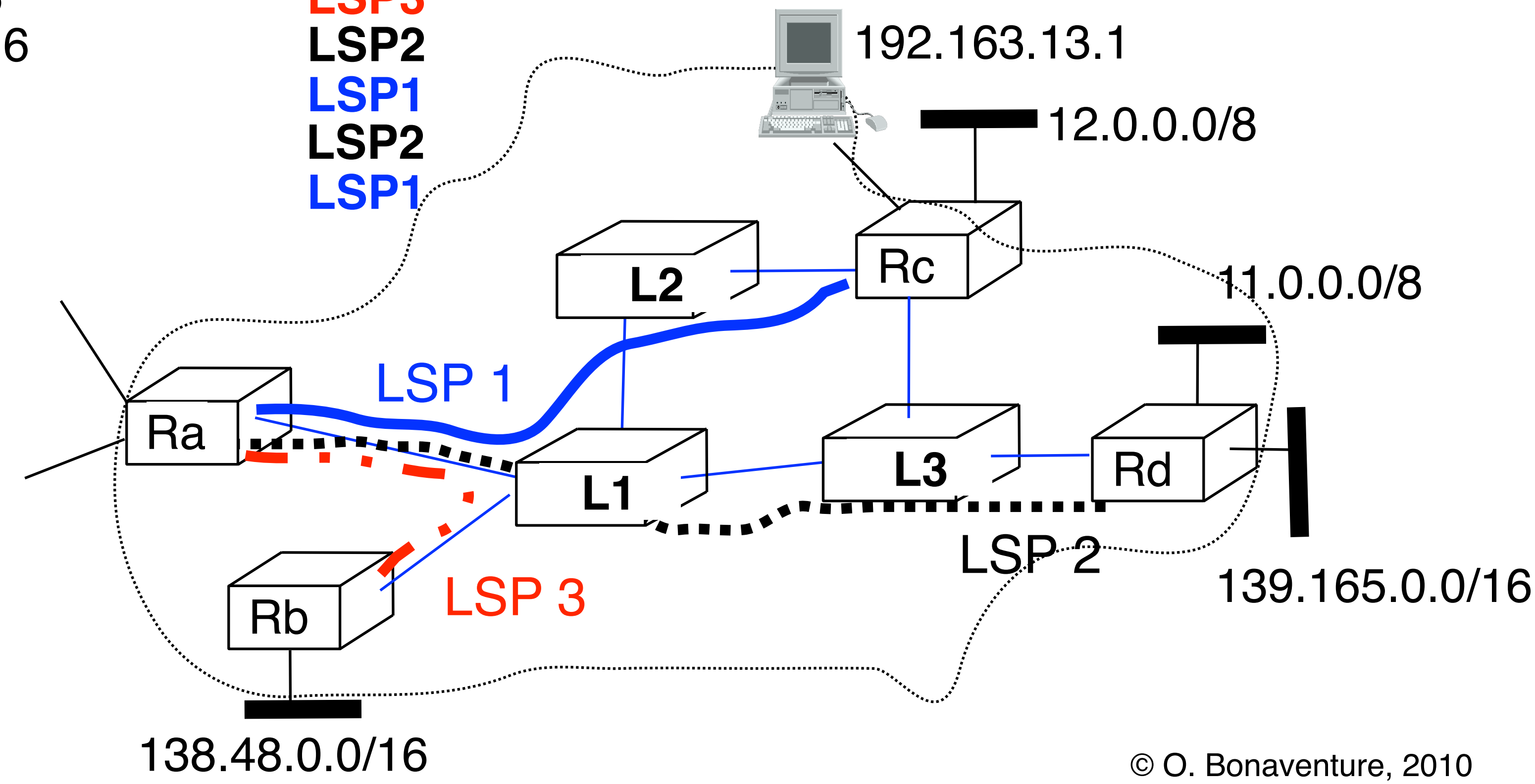
Behaviour of ingress edge LSR (2)

- Example

- LSP1
- Ingress=Ra, L1, L2, Egress=Rc
- LSP2
- Ingress=Ra, L1, L3, Egress=Rd
- LSP3
- Ingress=Ra, L1, Egress=Rb

Ra's Mapping table

IP dest prefix	LSP
138.48.0.0/16	**LSP3**
139.165.0.0/16	**LSP2**
12.0.0.0/8	**LSP1**
11.0.0.0/8	**LSP2**
192.163.13.1	**LSP1**

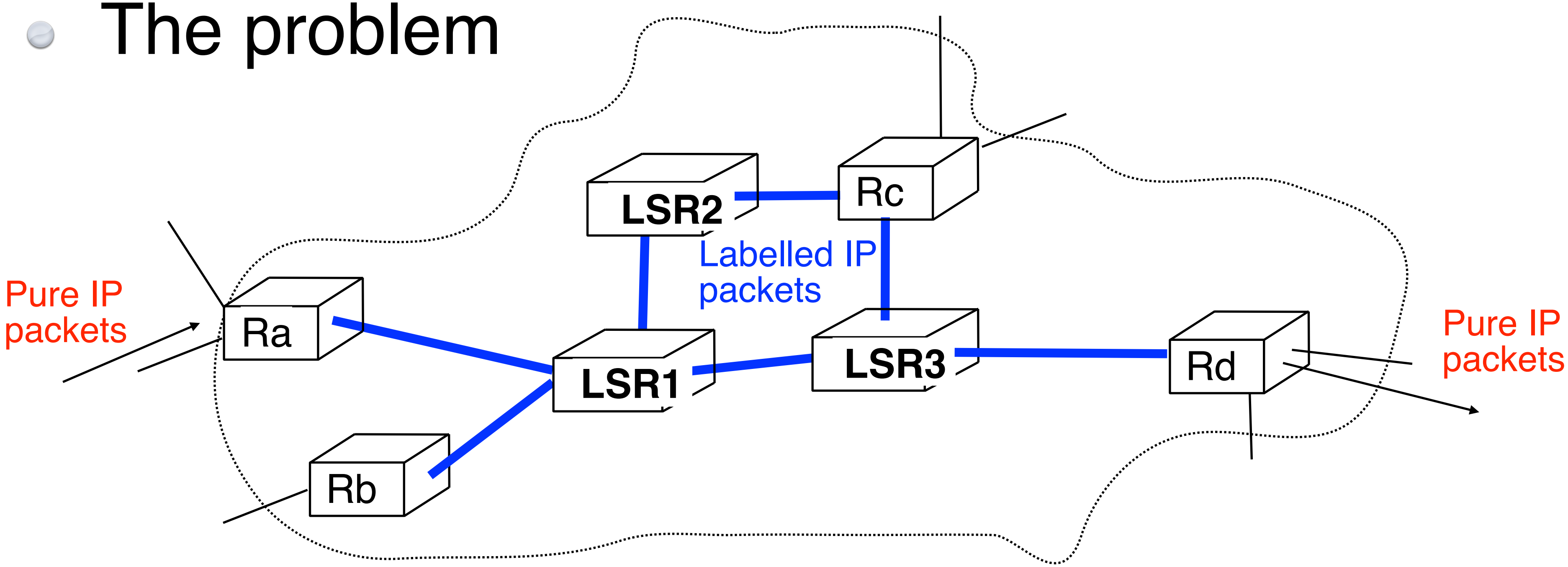


MPLS
MultiProtocol Label Switching

- Plan
  - Multiprotocol Label Switching
    - The label swapping forwarding paradigm
    - Integrating label swapping and IP
  - Utilisations of MPLS
    - Destination based packet forwarding
    - Simpler ISP backbones
    - Traffic engineering
    - QoS support
    - Fast restoration

Destination-based packet forwarding

- The problem

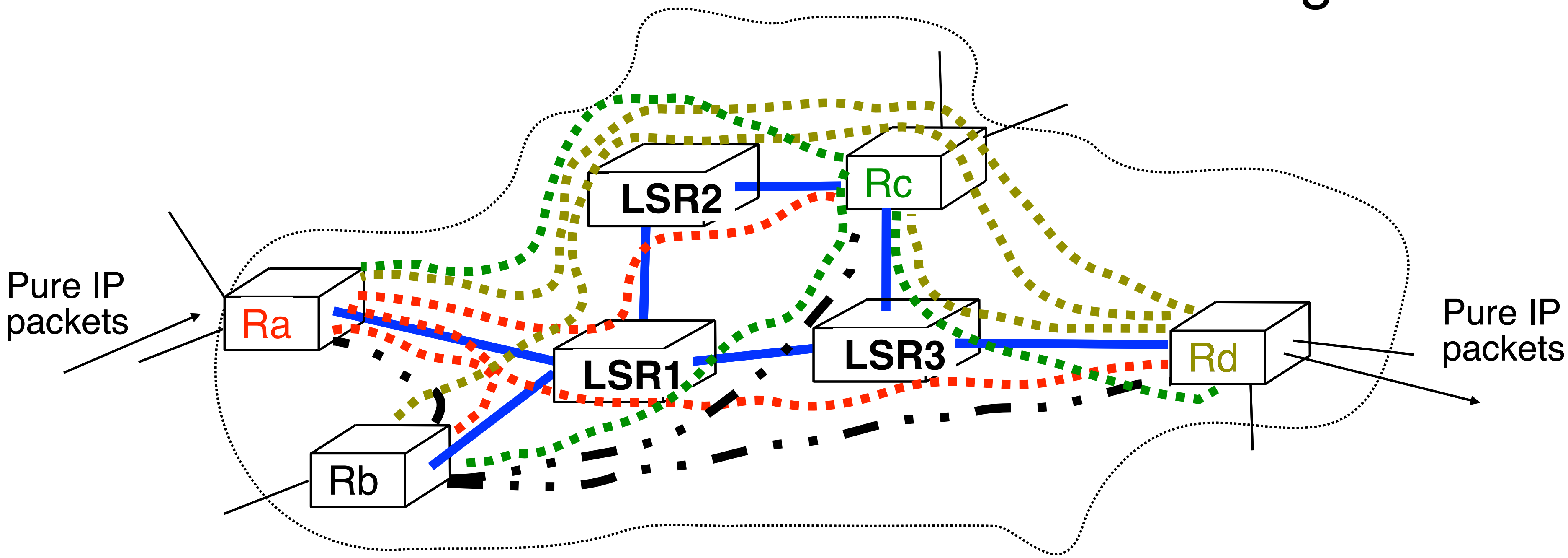


- How to provide transit service when
  - Edge LSRs are able to attach and remove labels
  - Edge LSR and Core LSRs run IP routing protocols and maintain IP routing tables
  - Core LSR can *only* forward labelled packets
    - Core LSR cannot route IP packets efficiently !

CNP3/2010.8.

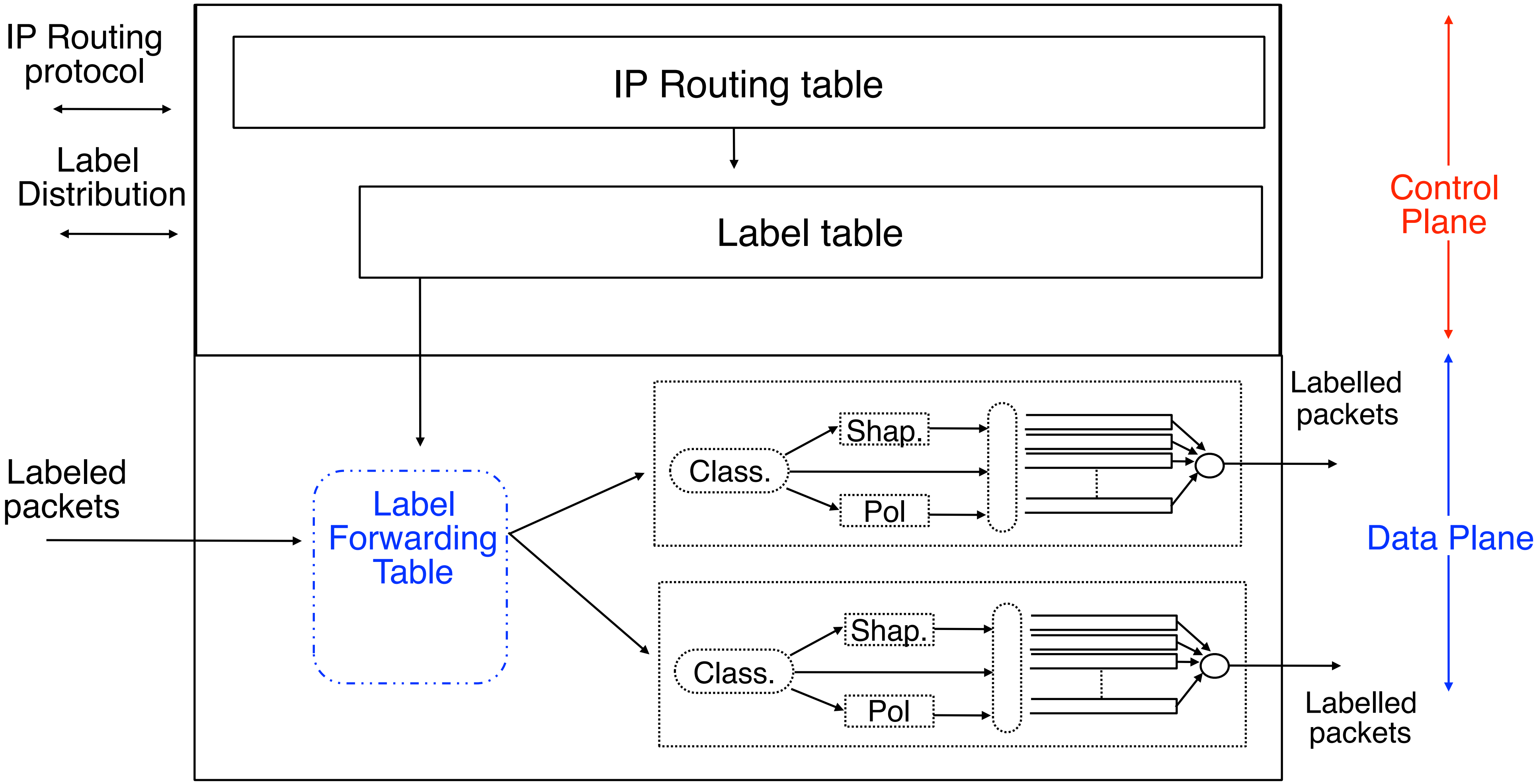
Destination-based packet forwarding (2)

- Manual solution
  - Create full mesh of LSPs between all edge LSRs

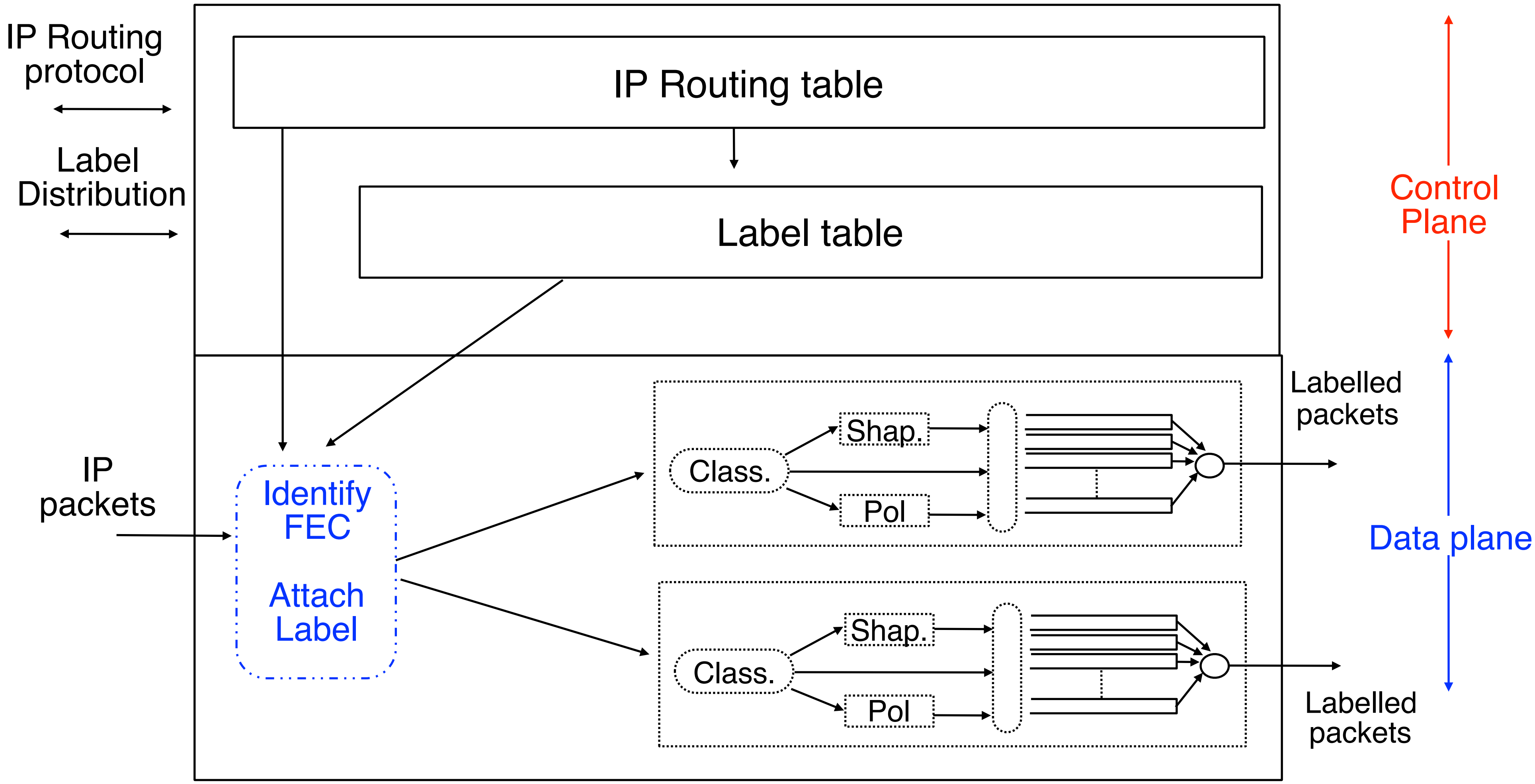


- Problems to be solved
  - N edge LSRs -> N*(N-1) unidirectional LSPs
    - How to automate LSP establishment ?
    - How to reduce the number of required LSPs ?

Architecture of a core LSR



Architecture of an ingress edge LSR

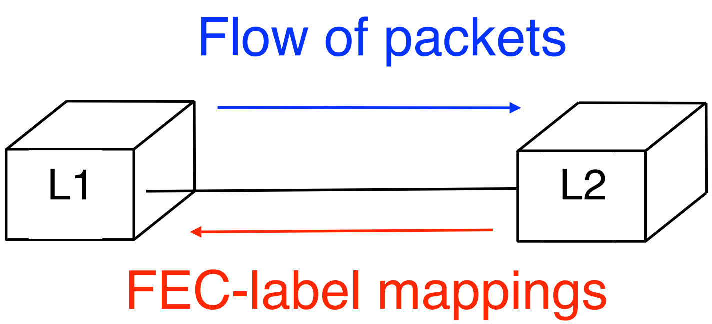


Distributing labels

- How to fill the label forwarding tables of all LSRs in a given network ?
  - Use a special protocol to distribute FEC-label mappings
    - LDP : Label Distribution Protocol
    - RSVP-TE : extensions to RSVP
  - Piggyback FEC-label mappings inside messages sent by routing protocol
    - possible if routing protocol is extensible
      - BGP can be easily modified to associate label with route
      - RIP cannot be used because its syntax is not extensible
  - Use global labels (Segment Routing)
    - link-state protocols (OSPF IS-IS) can distribute such labels

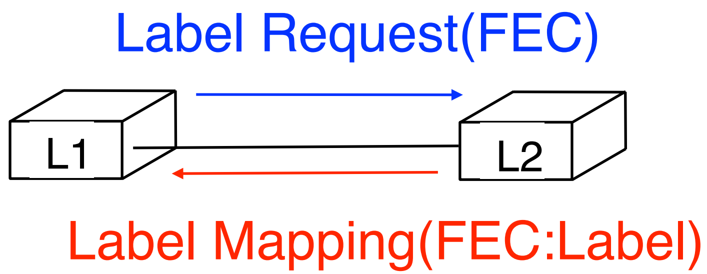
How to distribute labels ?

- Who determines the FEC-label mapping ?
  - packets are sent by upstream LSR towards downstream LSR
  - FEC-label mappings are sent by downstream LSR towards upstream LSR



Label distribution modes

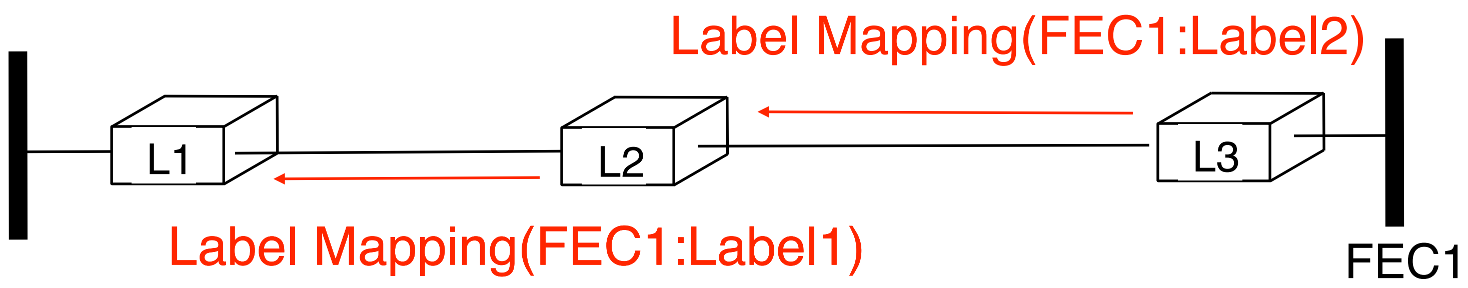
- When to distribute the FEC-label mapping?
  - Downstream on demand label distribution
    - upstream LSR requests and downstream allocates label



    - advantages
      - the FEC-label mappings are only distributed when needed
      - LSR only need to store the FEC-label mappings that are in use
    - drawback
      - when a next-hop fails, some time may elapse while a new FEC-label mapping is been requested from the new next-hop

Label distribution modes (2)

- When to distribute the FEC-label mapping ?
  - Unsolicited downstream label distribution
    - downstream LSR announces independently FEC-label mappings to upstream LSR



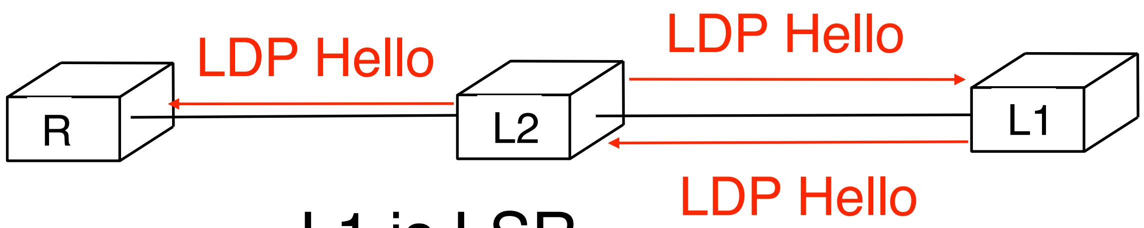
    - advantage
      - Each LSR can obtain several labels for each FEC
      - in case of failure, LSR can quickly switch from one label to another
    - drawback
      - Labels may not be distributed at the best time

Label Distribution Protocol

- LDP : Label Distribution Protocol
  - Designed to distribute FEC-labels mapping on a hop-by-hop basis inside network when labels cannot be distributed by routing protocols
  - Neighbour discovery over UDP
      - determine whether the neighbour is a LSR or a normal router
  - Distribution of FEC-label mappings over TCP
    - several modes of distribution are supported by LDP
    - we will only provide some examples of LDP
    - Main messages
      - Initialisation
        - establishment of a LDP session
      - Keepalive
        - used to verify that the LDP session is still up
      - Label mapping
        - used by LSR to announce a FEC-label mapping
      - Label withdrawal
        - Used by LSR to withdraw a previous FEC-label mapping

Neighbour discovery

- Principle
  - LSR periodically send LDP Hello packets to neighbor on "all routers" multicast address
    - LDP neighbour discovery uses UDP port 646
  - neighbours respond with LDP Hello if they are LSR



L1 is LSR
R is not LSR

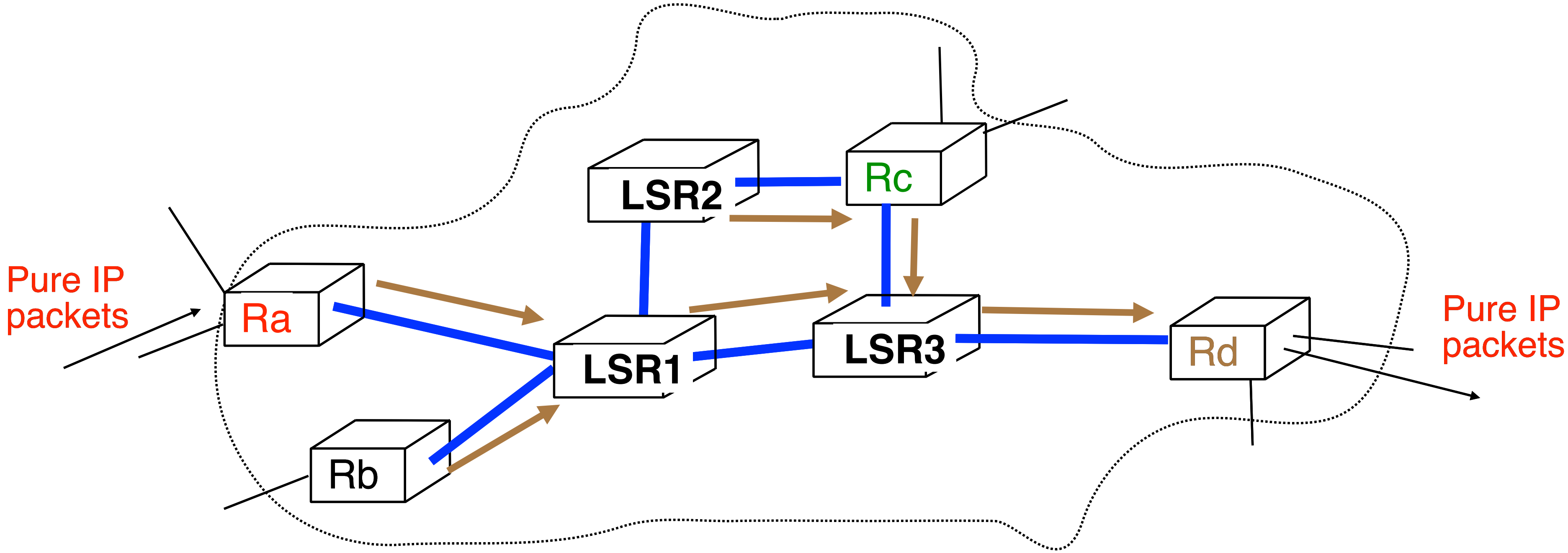
- LSR with highest IP address becomes active and establishes TCP connection for LDP on port 646
- LSR with lowest IP address becomes passive and waits the establishment of TCP connection for LDP session
  - TCP session established on port 646
  - LDP session establishment allows negotiation of options

LDP messages

- Initialisation
  - used during LDP session establishment to announce and negotiate options
- Keepalive
  - sent periodically in absence of other messages
- Label mapping
  - used by LSR to announce a FEC-label mapping
- Label withdrawal
  - used by LSR to withdraw a previous FEC-label mapping
- Label Release
  - used by LSR to indicate that it will not use a previously received FEC-label mapping
- Label Request
  - used by LSR to request a label for a specific FEC

Destination based forwarding

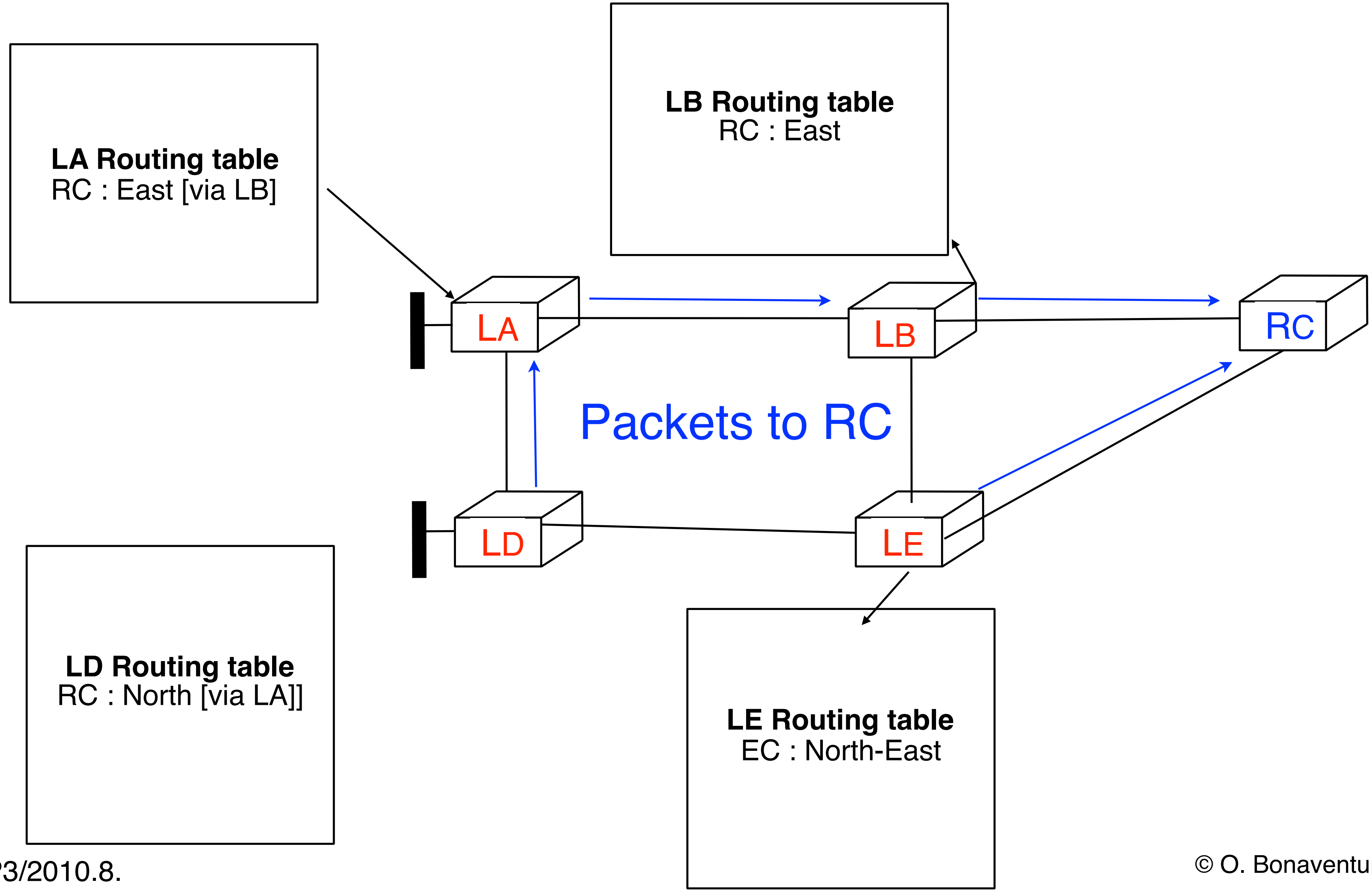
- Principle
  - Labelled packets should follow the same path through the network as if they were pure IP packets



- create a tree shaped LSP rooted on each egress LSR
  - similar to the way IP routing would forward packets
  - one tree per egress LSR, reduces total number of LSPs
- distribute the labels to build those trees

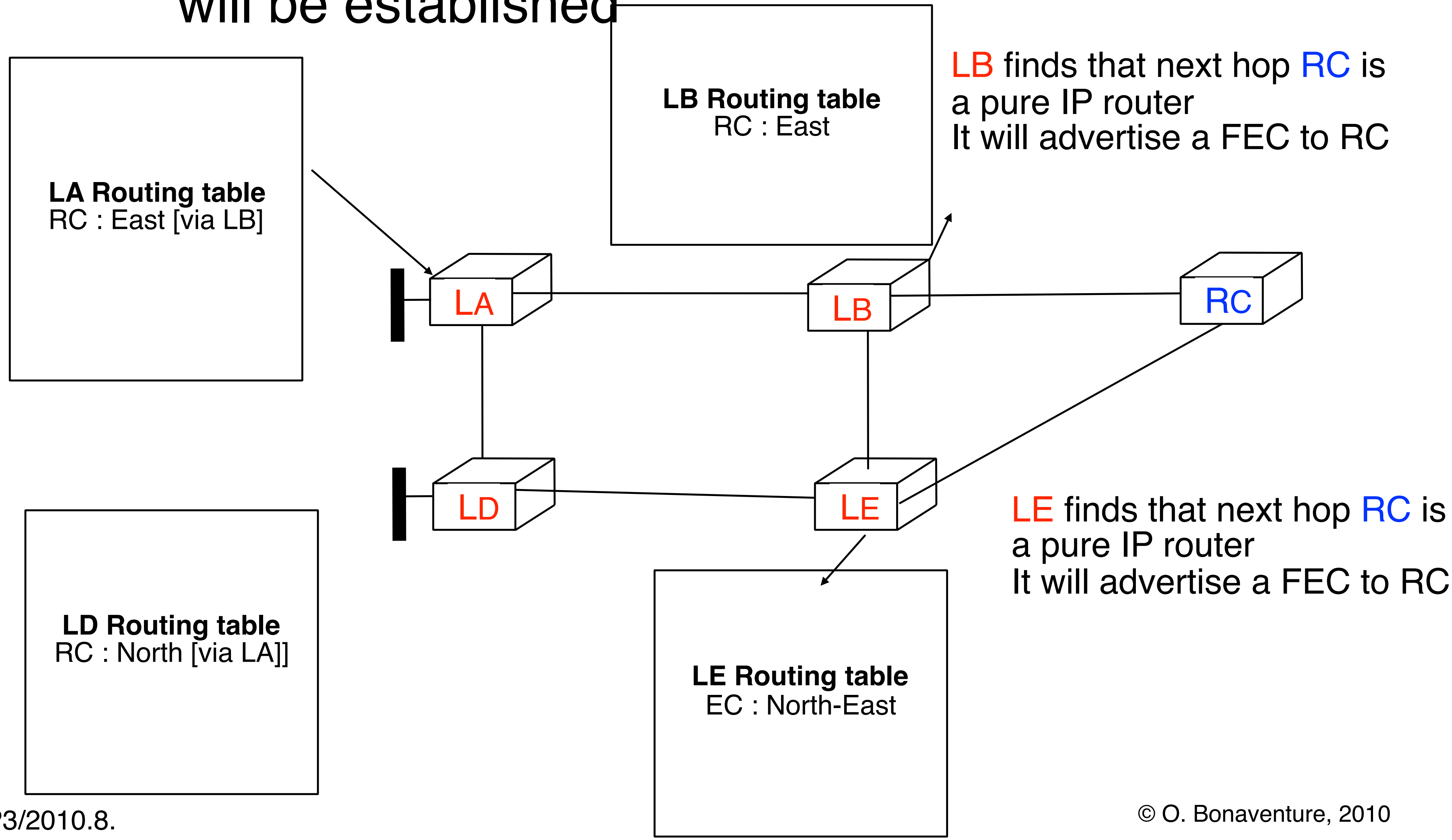
Destination based forwarding (2)

- How does LDP create the LSPs ?



Destination based forwarding (3)

- First step : choose destination for which a LSP will be established



Destination based forwarding (4)

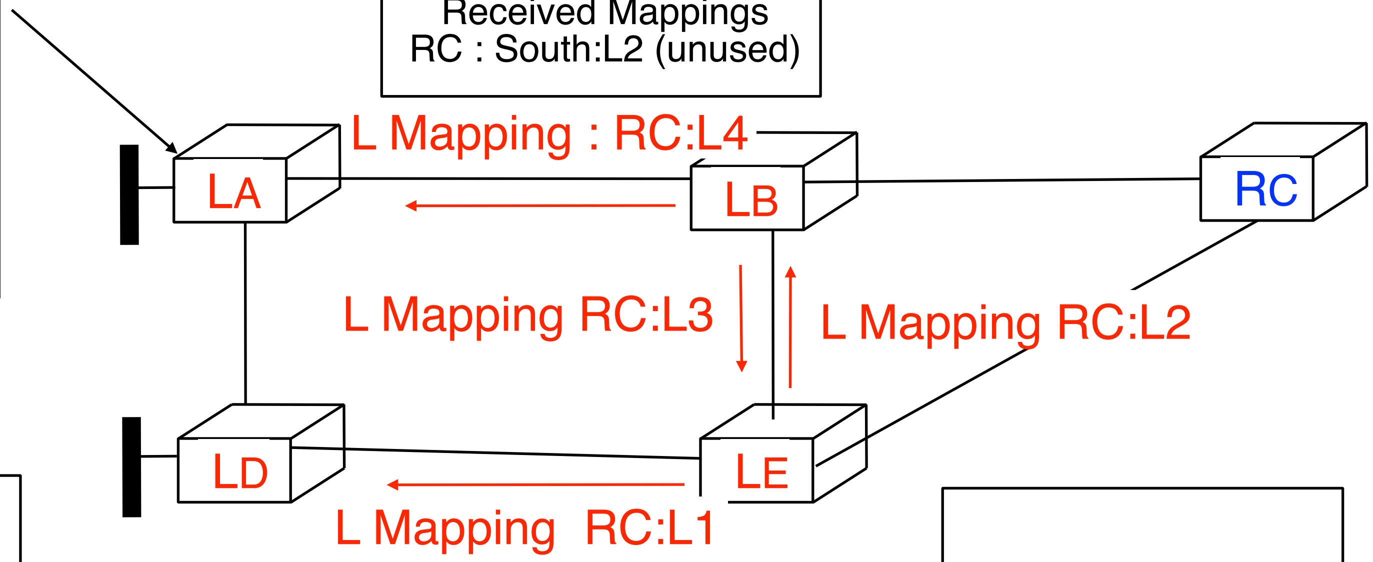
- Advertising the mappings at LB and LE

Label table
South:L3 -> East:POP
West:L4 -> East:POP

Received Mappings
RC : South:L2 (unused)

Label table
-
FEC table
RC -> PUSH L4, East

Received Mappings
RC : East:L4 (chosen)

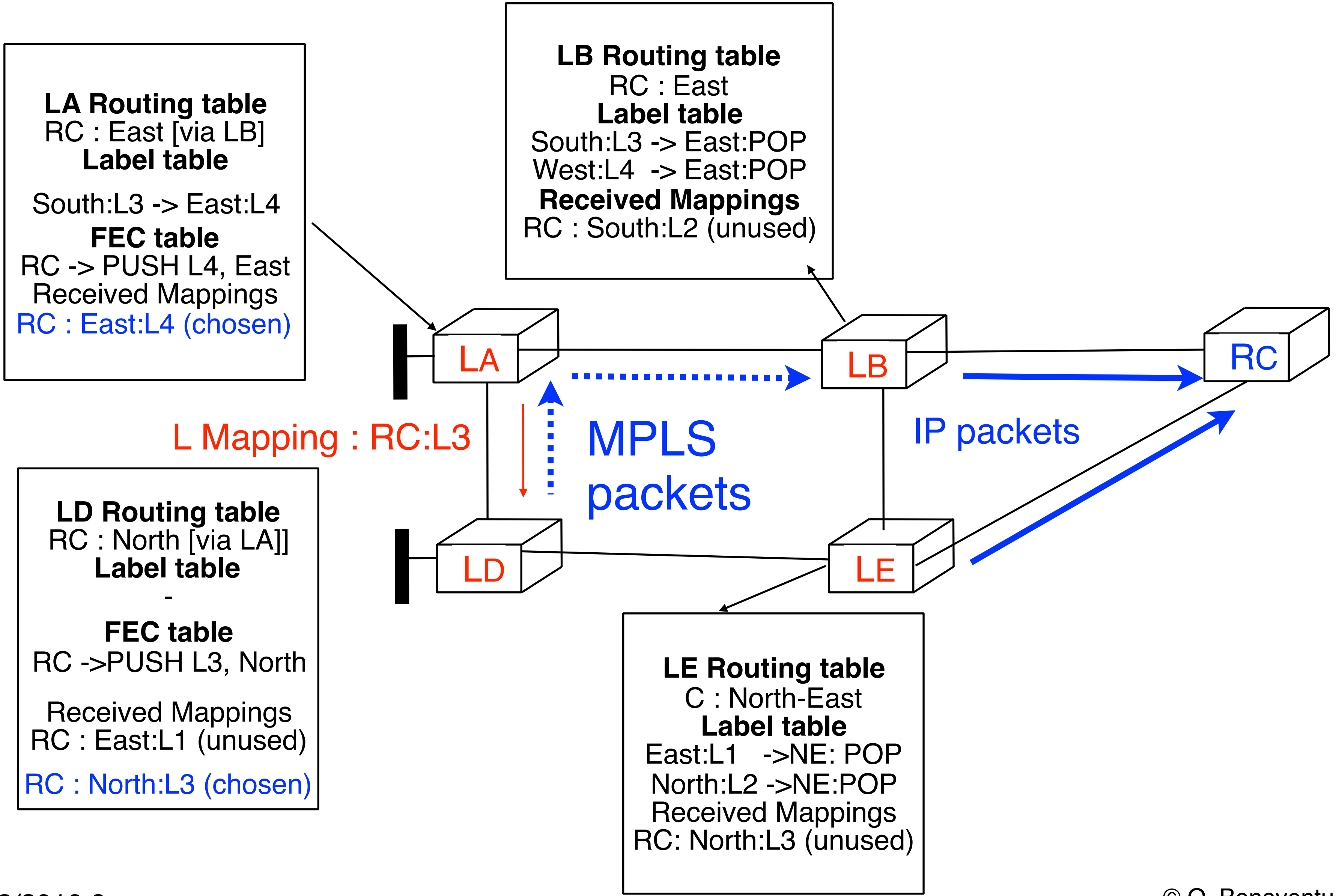


Label table
-
FEC table
-
Received Mappings
RC : East:L1 (unused)

Label table
East:L1 ->NE: POP
North:L2 ->NE:POP

Received Mappings
RC: North:L3 (unused)

Destination based forwarding (5)

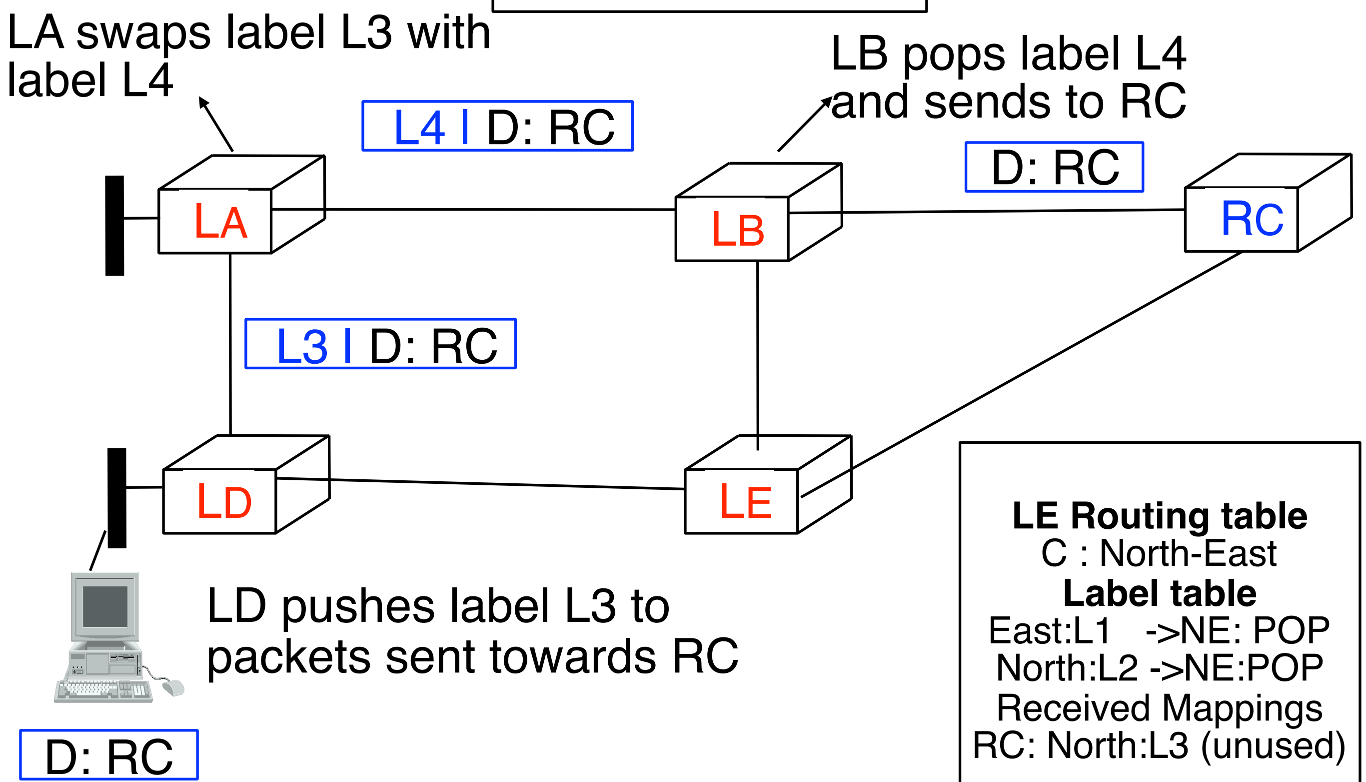


Destination based forwarding (6)

- Packet flow

LB Routing table
RC : East
Label table
South:L3 -> East:POP
West:L4 -> East:POP
Received Mappings
RC : South:L2 (unused)

LA Routing table
RC : East [via LB]
Label table
South:L3 -> East:L4
FEC table
RC -> PUSH L4, East
Received Mappings
RC : East:L4 (chosen)



LD Routing table
RC : North [via LA]]
Label table
-
FEC table
RC ->PUSH L3, North
Received Mappings
RC : East:L1 (unused)
RC : North:L3 (chosen)

LE Routing table
C : North-East
Label table
East:L1 ->NE: POP
North:L2 ->NE:POP
Received Mappings
RC: North:L3 (unused)

Destination based forwarding (7)

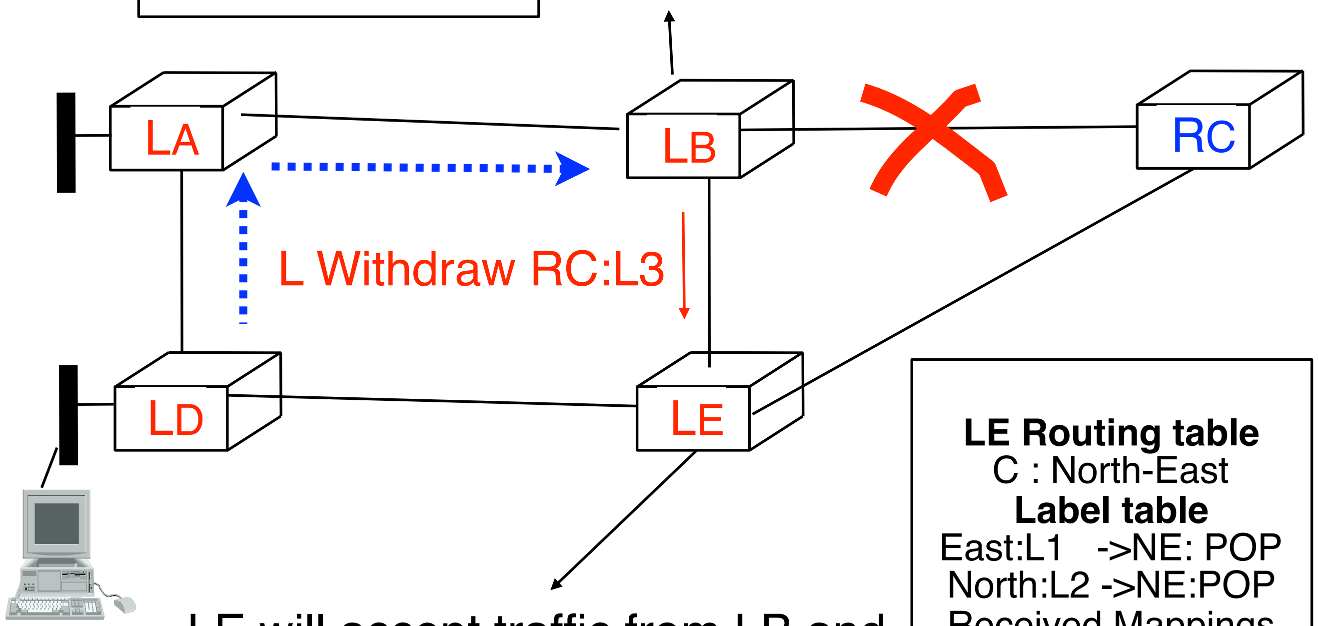
- How to deal with link failures ?

LA Routing table
RC : East [via LB]
Label table
South:L3 -> East:L4
FEC table
RC -> PUSH L4, East
Received Mappings
RC : East:L4 (chosen)

LB Routing table
RC : East (failed)

Label table
South:L3 -> East:POP
West:L4 -> South:L2
Received Mappings
RC : South:L2

LB notices the link failure
Since LB received a mapping from LE, it may immediately update its label table to "reroute" the traffic towards LE



LD Routing table
RC : North [via LA]]
Label table
-
FEC table
RC ->PUSH L3, North
Received Mappings
RC : East:L1
RC : North:L3 (chosen)

LE will accept traffic from LB and will remove the RC: North:L3 label from its received mappings

LE Routing table
C : North-East
Label table
East:L1 ->NE: POP
North:L2 ->NE:POP
Received Mappings
~~RC : North:L3~~

MPLS and transient loops

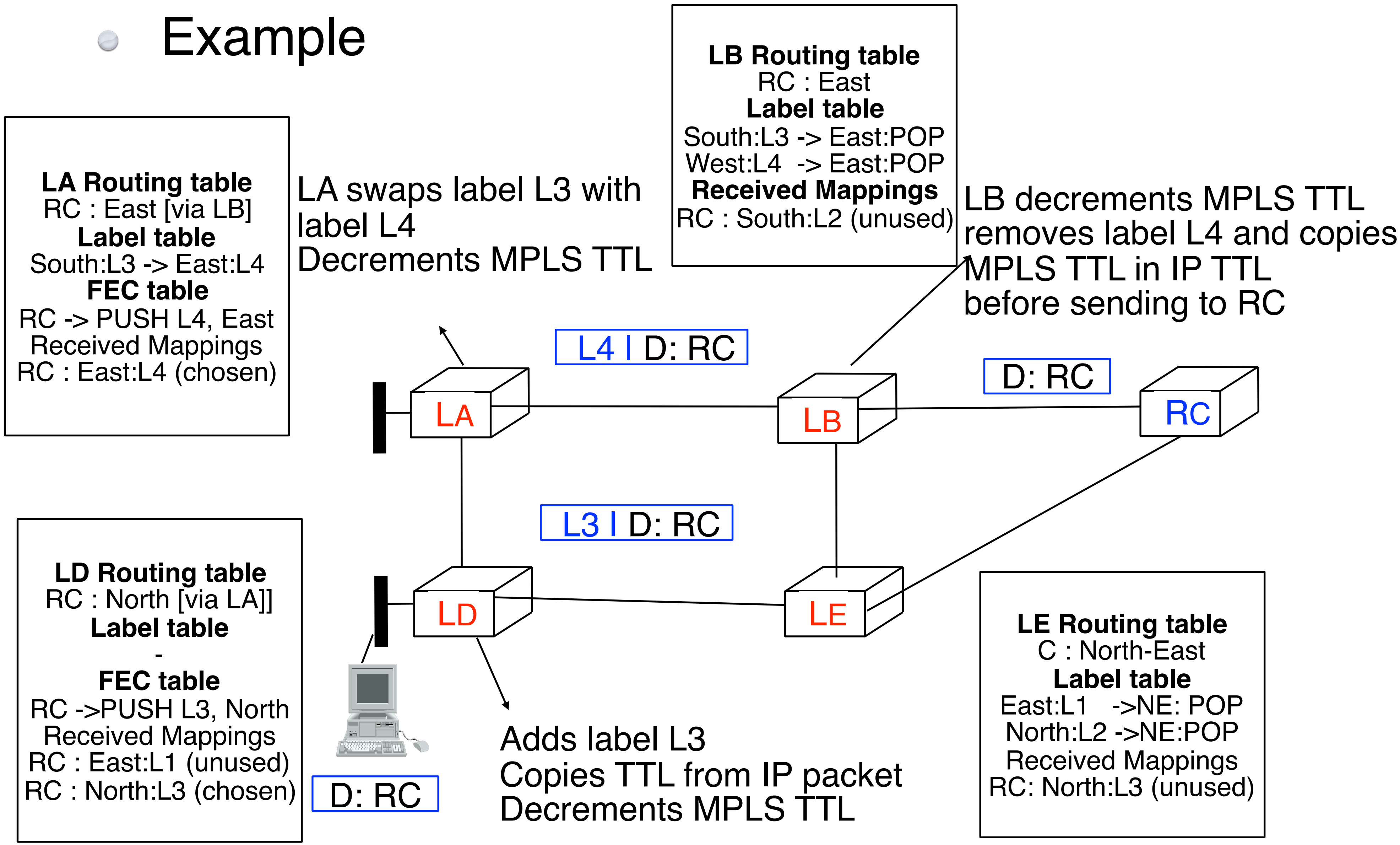
- What if routing has created a transient loop while LSP is being established ?
  - LSP could be looped inside network
- How to recover from looped LSPs ?
  - Discard looped packets as in IP
    - TTL field in MPLS header

```
                    1                   2                   3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+---------------------------------------+-----+-+---------------+
|                 Label                 |     |S|      TTL      |
+---------------------------------------+-----+-+---------------+
```

    - not suitable for technologies with their own header (ATM)
  - Prevent loops inside LDP
    - indicate the entire path of LSP in LDP Label request and LDP label mapping messages

Utilisation of the MPLS TTL

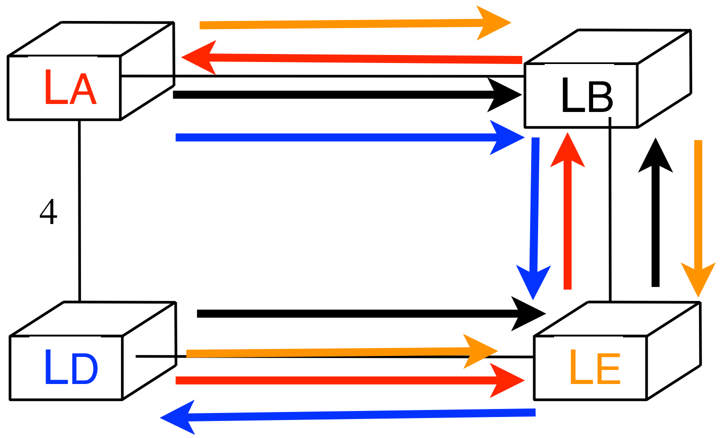
- Example



LDP in practice

- How do network operators use LDP ?
  - Most common deployment is to create a full mesh of LSPs among all LSRs so that each LSR is able to send MPLS packets to any LSR
  - LSR usually advertises a label for its loopback address

- What will be the LSPs in the network below ?

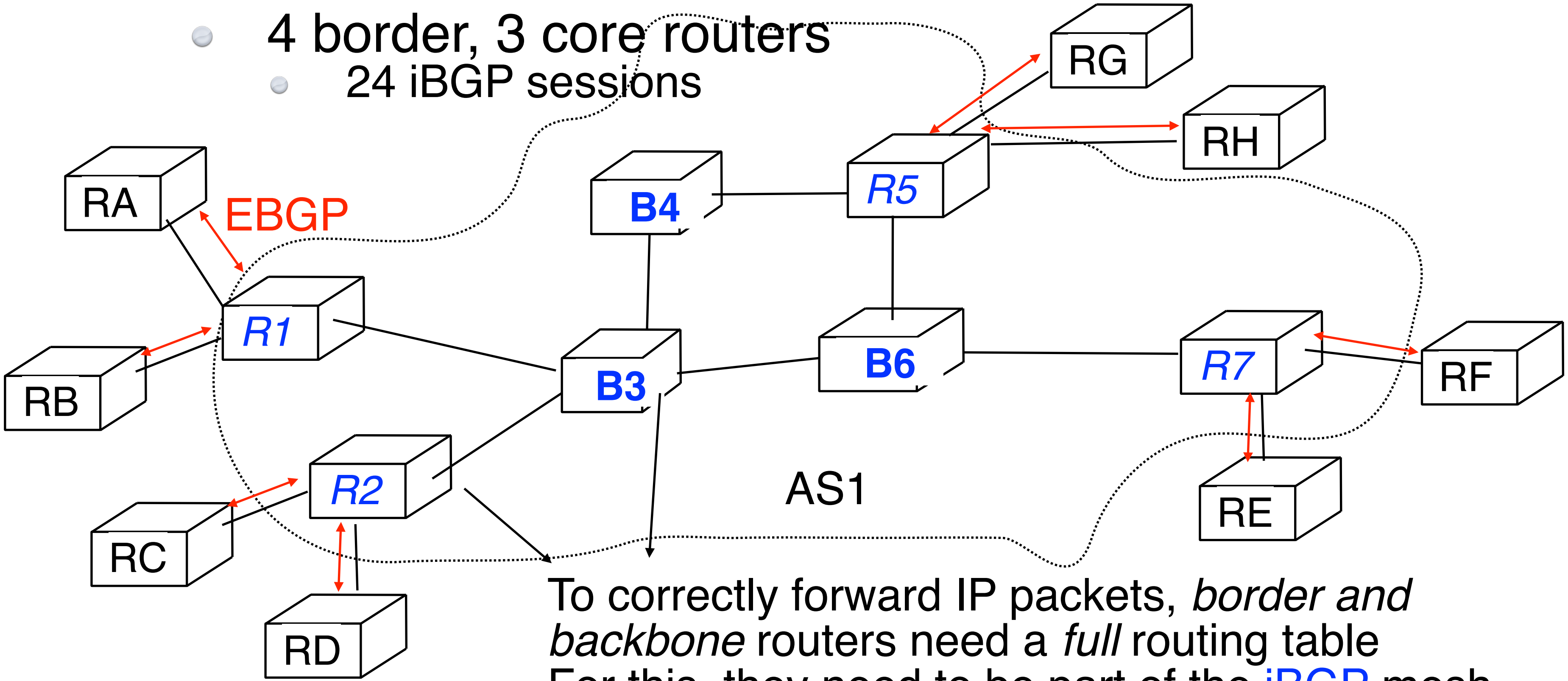


MPLS
MultiProtocol Label Switching

- Outline
  - Multiprotocol Label Switching
    - The label swapping forwarding paradigm
    - Integrating label swapping and IP
  - Utilisation of MPLS
    - Destination based packet forwarding
    - Simpler ISP backbones
    - Traffic engineering
    - QoS support
    - Fast restoration

MPLS in large ISP networks

- Pure IP-based ISP network
  - eBGP on border routers
    - current full BGP Internet routing table
      - +-400k active routes
  - iBGP full mesh
    - 4 border, 3 core routers
      - 24 iBGP sessions

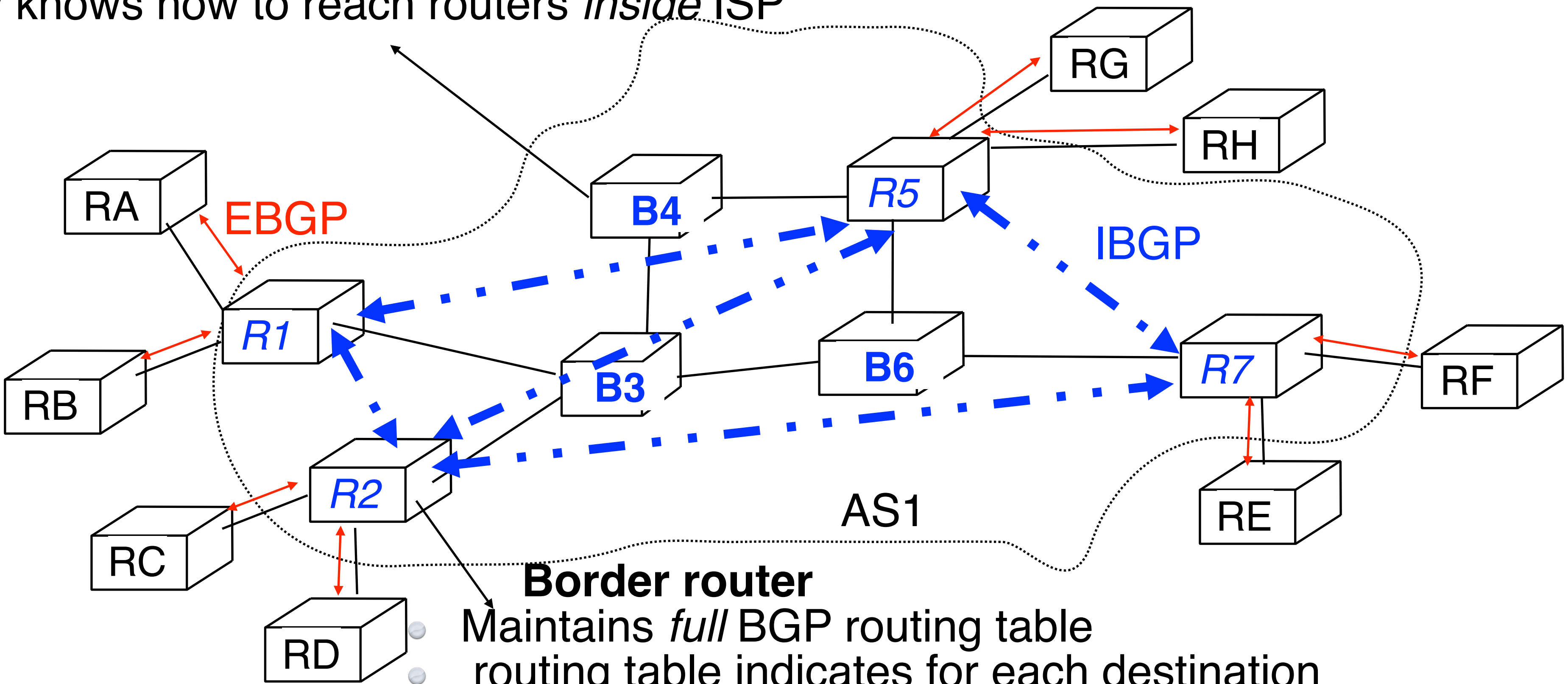


To correctly forward IP packets, *border and backbone* routers need a *full* routing table
For this, they need to be part of the iBGP mesh

MPLS in large ISP networks (2)

- BGP free ISP backbone

Backbone router
Maintains *internal* routing table of ISP network
only knows how to reach routers *inside* ISP

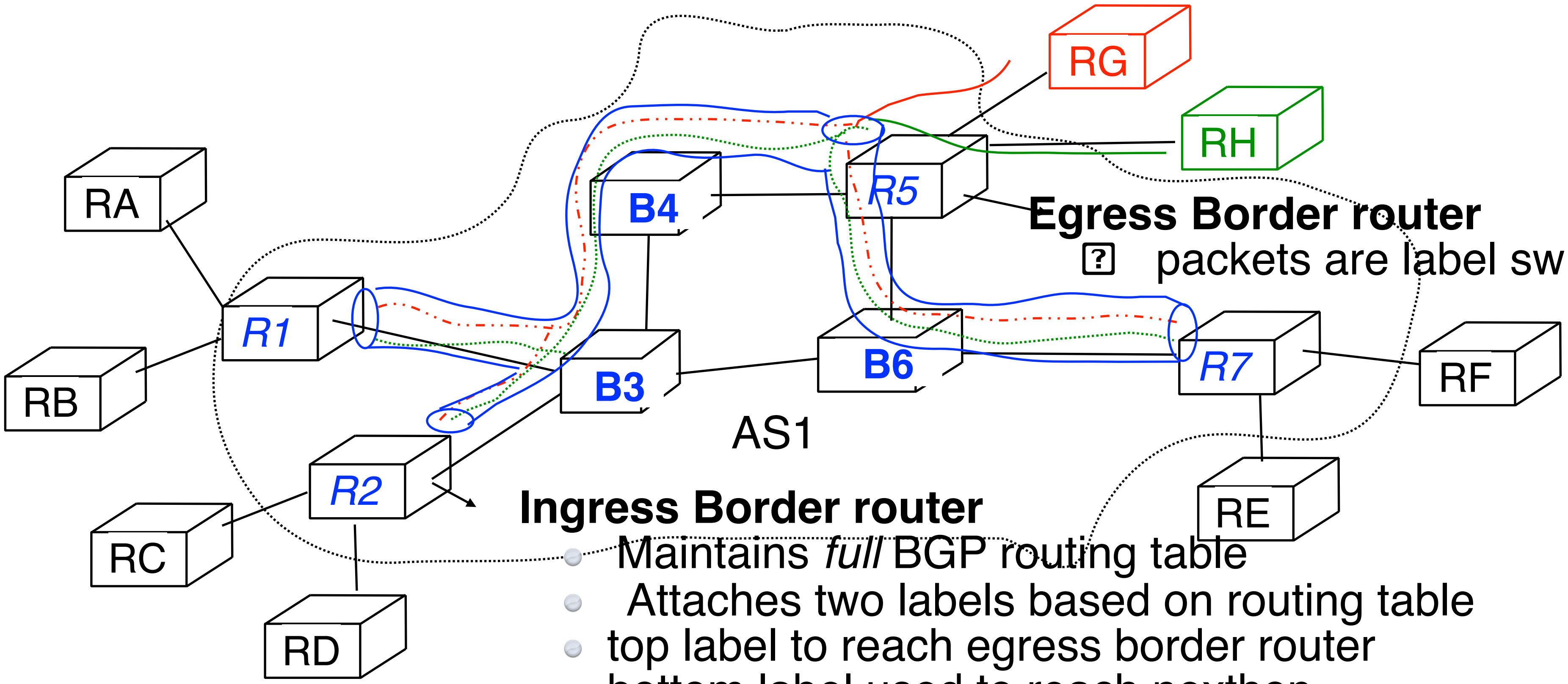


Border router

- Maintains *full* BGP routing table
- routing table indicates for each destination
- AS path towards destination
- IP address of next-hop to reach destination

MPLS in large ISP networks (3)

- Principle of the solution
  - Use a hierarchy of labels
    - top label is used to reach egress border router (blue LSP)
    - second label is used to reach eBGP peer (red/green LSP)



Egress Border router

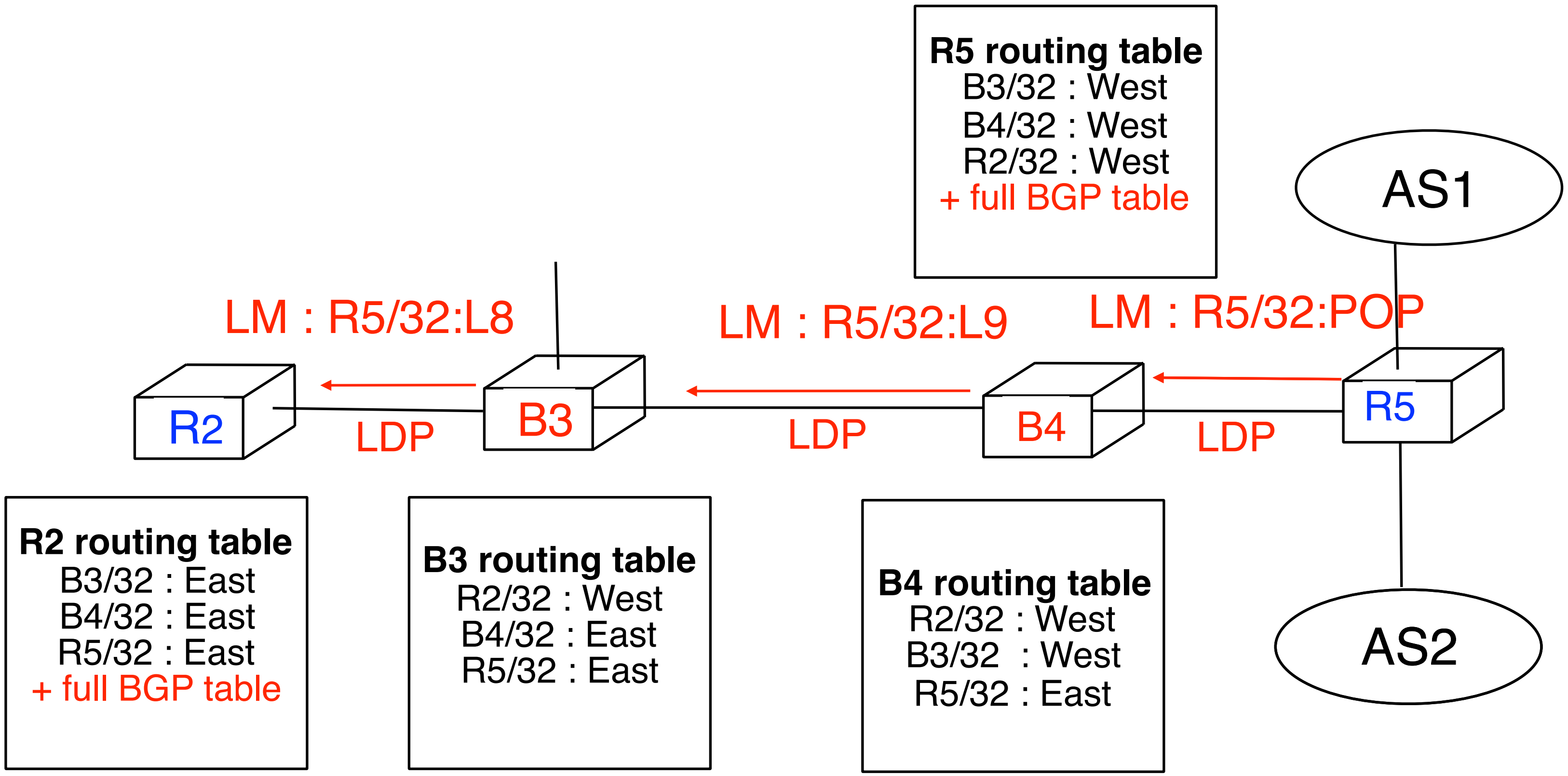
- packets are label switched

Ingress Border router

- Maintains *full* BGP routing table
- Attaches two labels based on routing table
- top label to reach egress border router
- bottom label used to reach nexthop

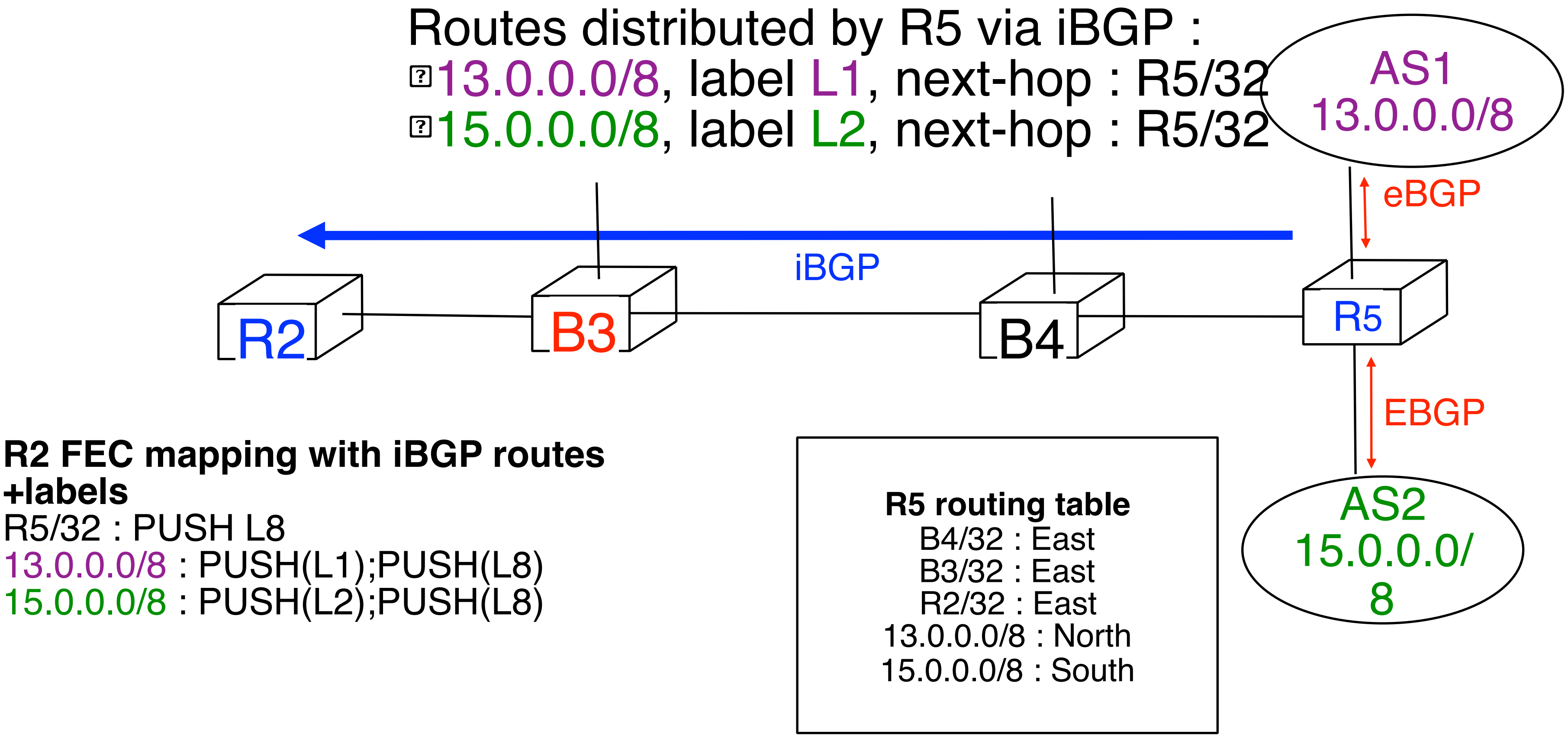
MPLS in large ISP networks (4)

- How to distribute the two labels ?
  - LDP allows to distribute the label to reach the egress BGP router

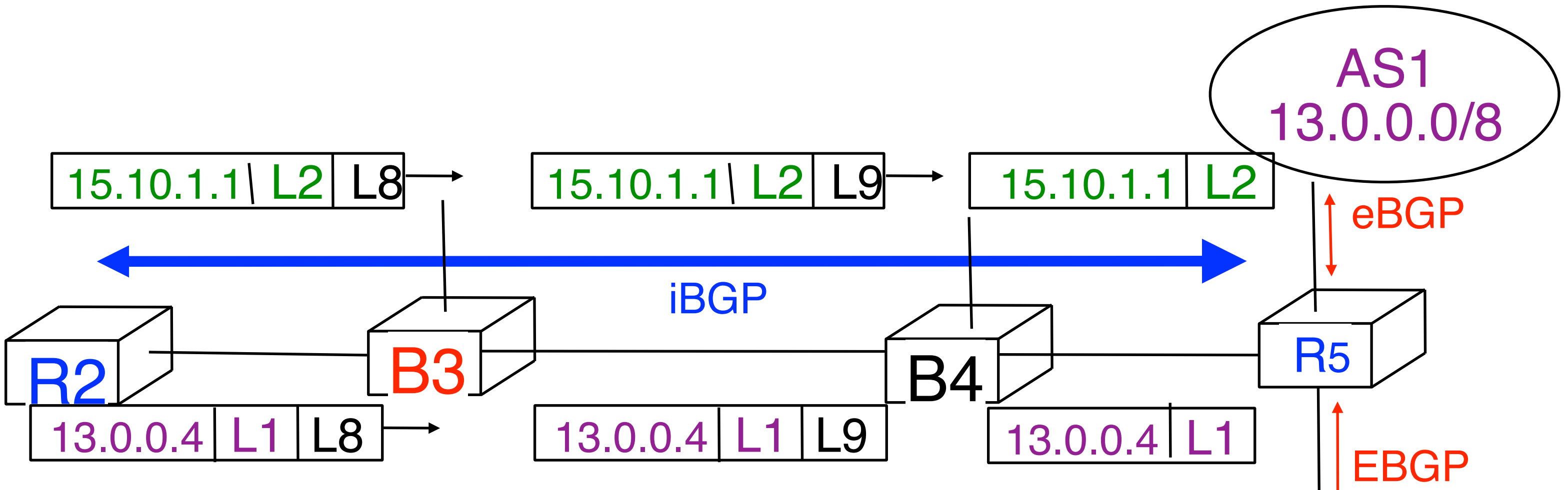


MPLS in large ISP networks (5)

- How to distribute the two labels ?
  - BGP allows to distribute the label associated to each prefix



MPLS in large ISP networks (6)



R2 FEC mapping with iBGP routes +labels
R2/32 : PUSH L8
13.0.0.0/8 : PUSH(L1);PUSH(L8)
15.0.0.0/8 : PUSH(L2);PUSH(L8)

B4 routing table
B3/32 : West
R2/32 : West
R5/32 : East

Label table
West:L9 -> East:POP

R2 routing table
B4/32 : East
B4/32 : East
R5/32 : East

LDP FEC mapping
R2/32 : PUSH L8

B4 routing table
R2/32 : West
B4/32 : East
R5/32 : East

Label table
West:L8 -> East:L9

R5 routing table
B4/32 : East
B3/32 : East
R2/32 : East
13.0.0.0/8 : North
15.0.0.0/8 : South

R5 Label table
West:L3 -> Local:POP
L1 -> North:POP
L2 -> South:POP